
Design of Analog CMOS Integrated Circuits-I

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Xi'an Jiaotong University
Hong Zhang (张 鸿)



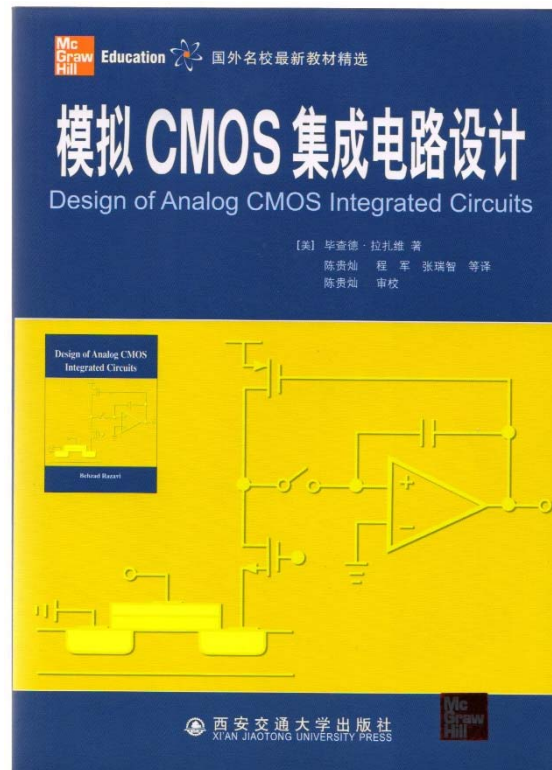
About Your Instructor

- Been with the Dept. of Microelectronics since 2008.
(AP in 2011, Professor in 2018)
- Visiting Professor in the ECE Dept., University of Toronto, 2016~2017
- Ph. D, Xi'an Jiaotong University 2008
 - Radio Frequency IC design: Ultra-WideBand (UWB) Transceiver.
- Current Research Area:
 - Low-power A/D converters
 - Bio-medical circuits and systems
 - Battery management IC design
- Office
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 - Email: hongzhang@xjtu.edu.cn

Course Basics (1)

- Required Textbook

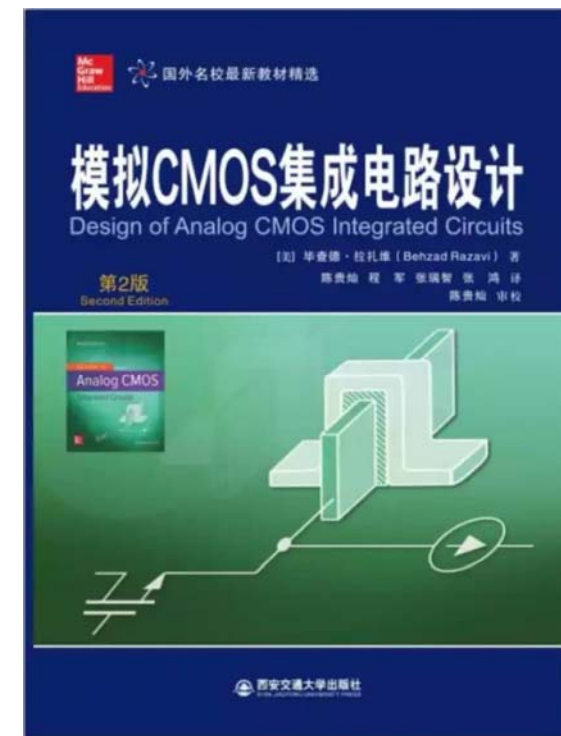
1st edition



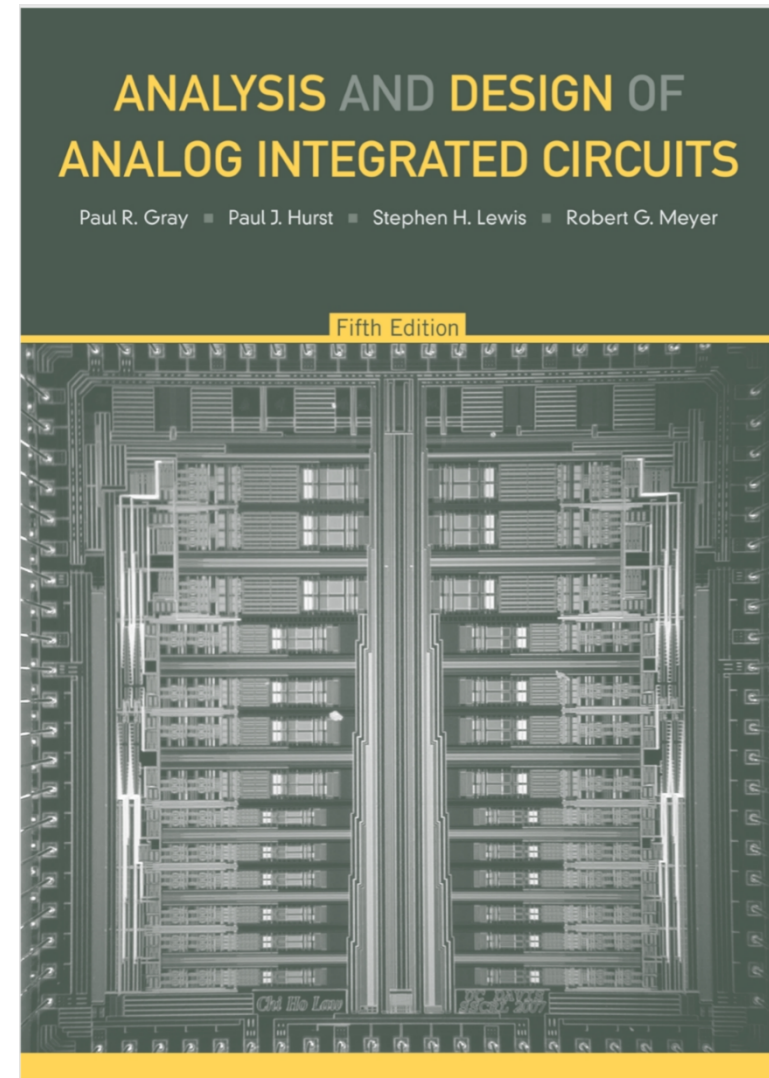
- Greater emphasis on modern CMOS technology
- Feedback system
- 100+ new examples



2nd edition



- Recommended texture book
 - *Analysis and Design of Analog Integrated Circuits*, 5th Edition, Gray, Hurst, Lewis and Meyer, Wiley, 2009.



Course Basics (2)

- Course prerequisites
 - Semiconductor Physics
 - Basic device physics and models
 - PN junctions, MOSFETs, BJTs
 - Basic linear systems
 - Frequency response, Poles, Zeros

Assignments

- Attendance (guaranteed by the teaching system)
- Homeworks (20%)
- Projects using Hspice (10~20%)
 - 1 ~ 2 times
- Final Exam (60~70%)
- Important note:
 - You can ask any question before the exam, but never get to me after the exam.

Teaching Assistant Information

- TBD

Other information

- Course Website: <http://syxt.xjtu.edu.cn>
 - Download resources including lecture notes, electronic books, Hspice software, example netlists, course videos, etc.
 - Discuss questions in the online-forums !!
 - Hand in assignments
- Online tools:
 - “长江雨课堂” for quizzes in the class

Chapter 1

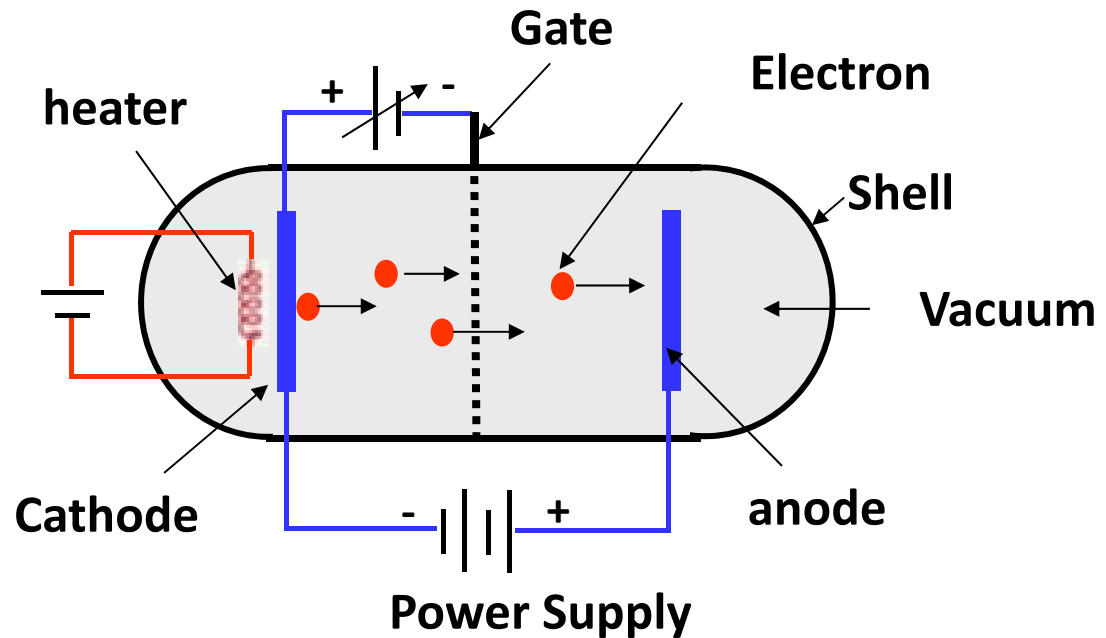
Introduction to Analog Design



History of IC

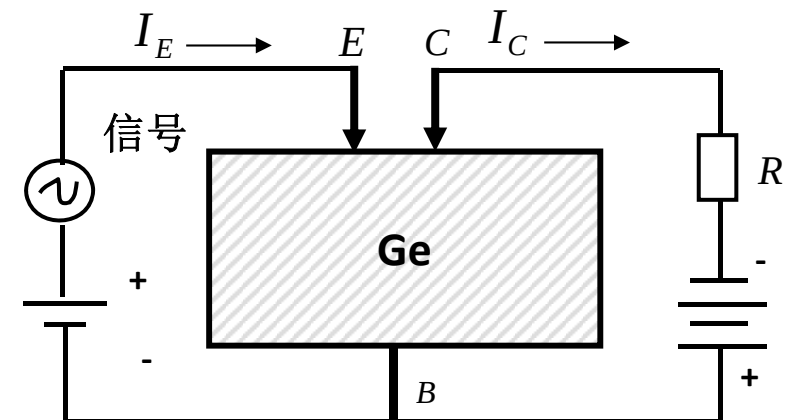
- 1906, Vacuum Tube: Current of anode controlled by Gate Voltage.
 - Amplifier or switch
 - Beginning of the electronics era.
 - Very large size
 - First computer , 18000 tubes, 30 ton, 90m³,

Vacuum Tube
1906



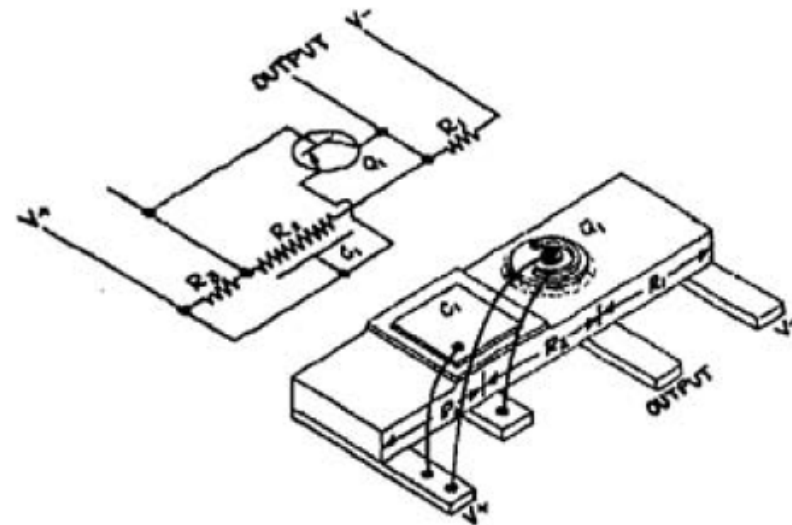
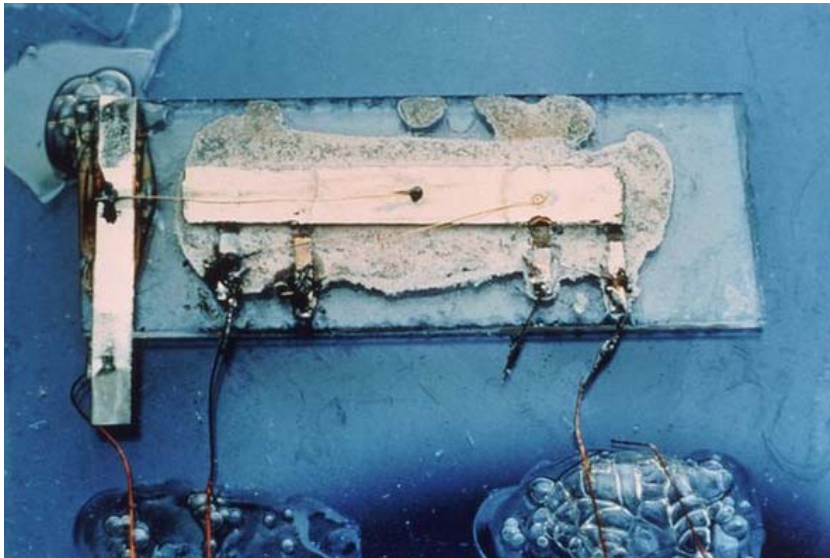
History of IC

- 1947, the first transistor invented by Shockly
- Semiconductor era opening
 - Nobel Prize, 1956



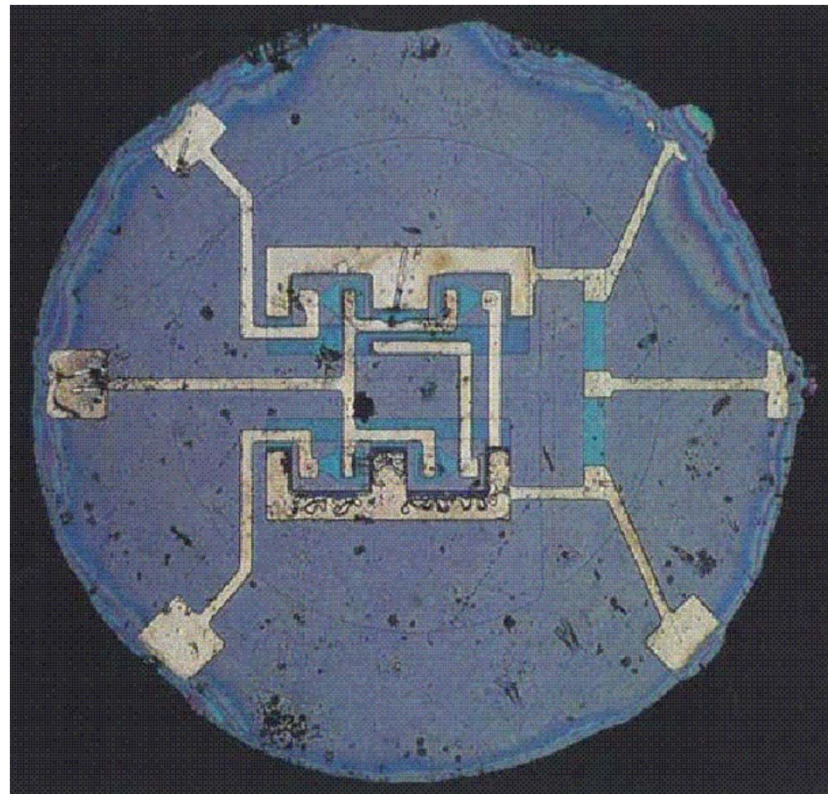
History of IC

- 1958, first Integrated Circuit (IC), an oscillator
 - Invented by Kilby, TI
 - Nobel Prize, 2000



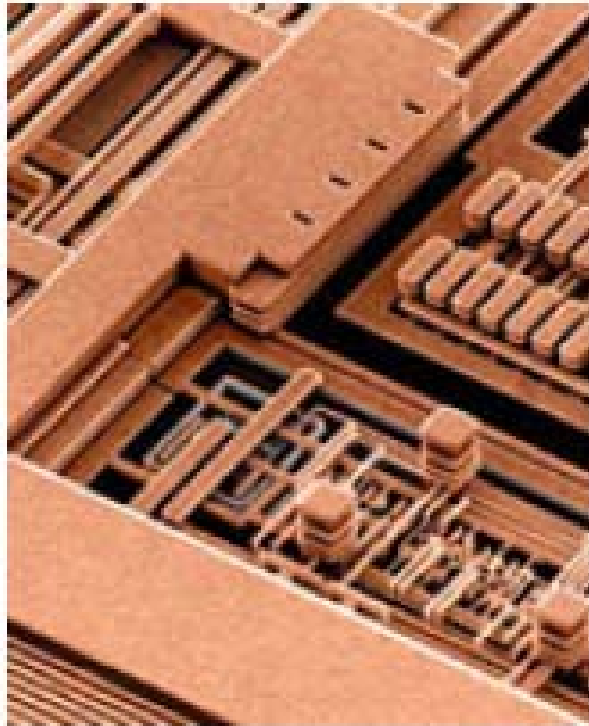
History of IC

- 1961, first commercial IC, 4 transistors and 2 resistors
 - Invented by Noyce (1927~1990)

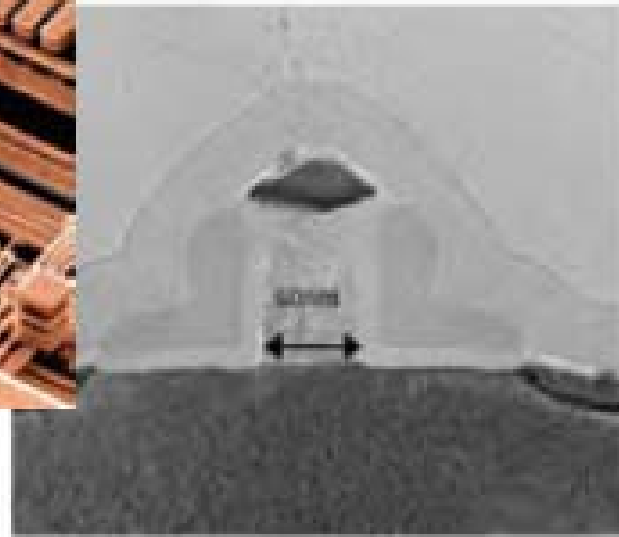


Modern IC: based on silicon

- Multi-layer metal on a semiconductor substrate
- Very high level of integration



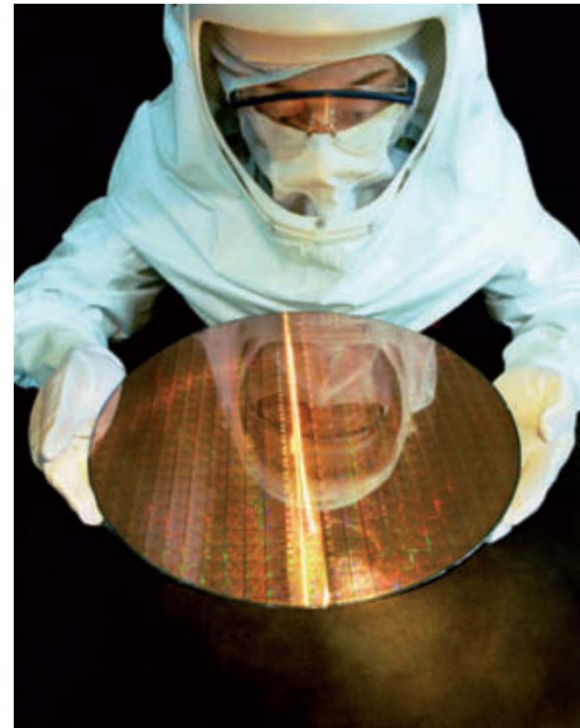
Modern
CMOS



Modern IC: Based on Silicon wafer



**Silicon ingot, 12-inch diameter,
1-m long**

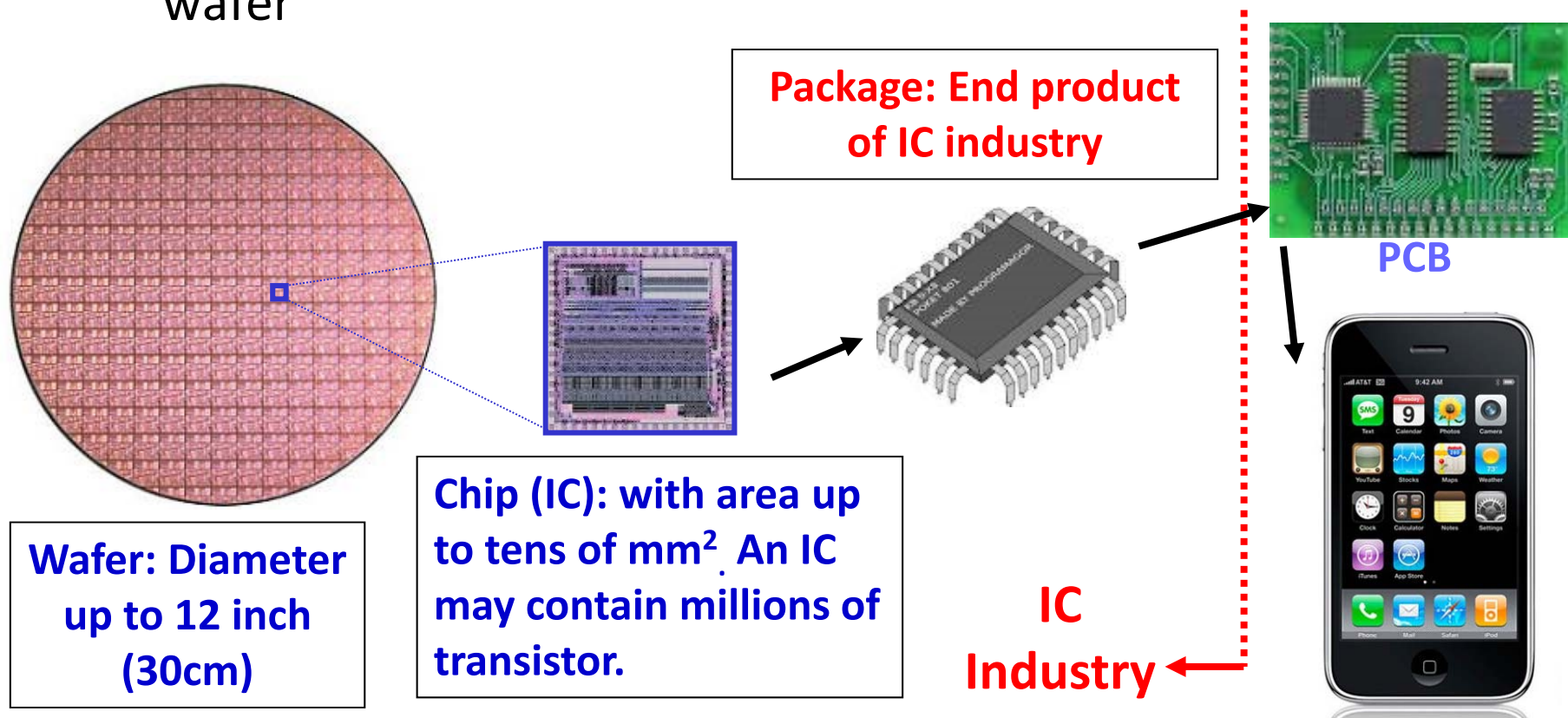


**Silicon wafer, 12-inch diameter,
0.75-mm thick**



Modern Semiconductor Industry

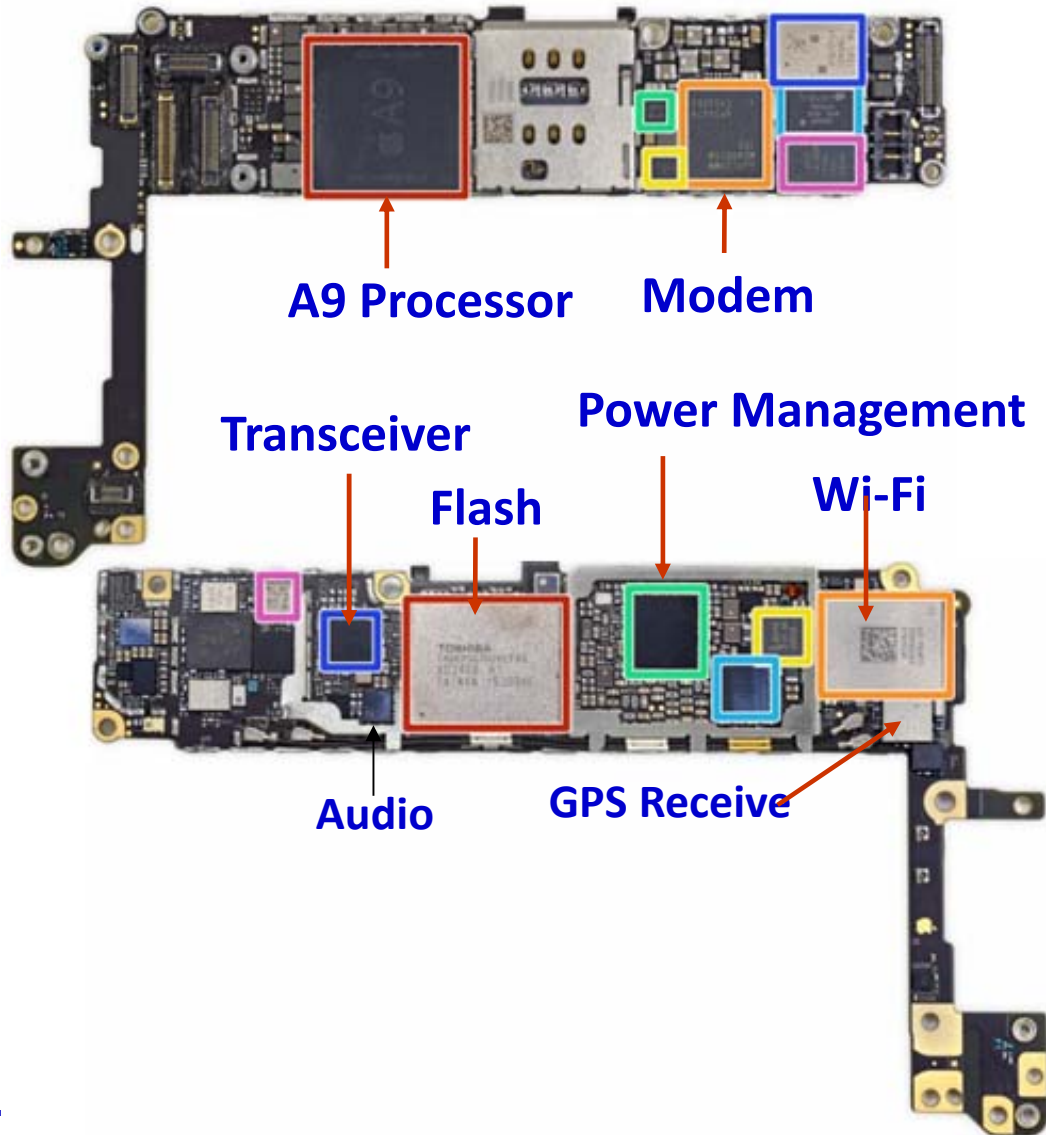
- The Goal of Semiconductor industry:
 - Producing Chips (or Integrated Circuits, IC) based on a silicon wafer

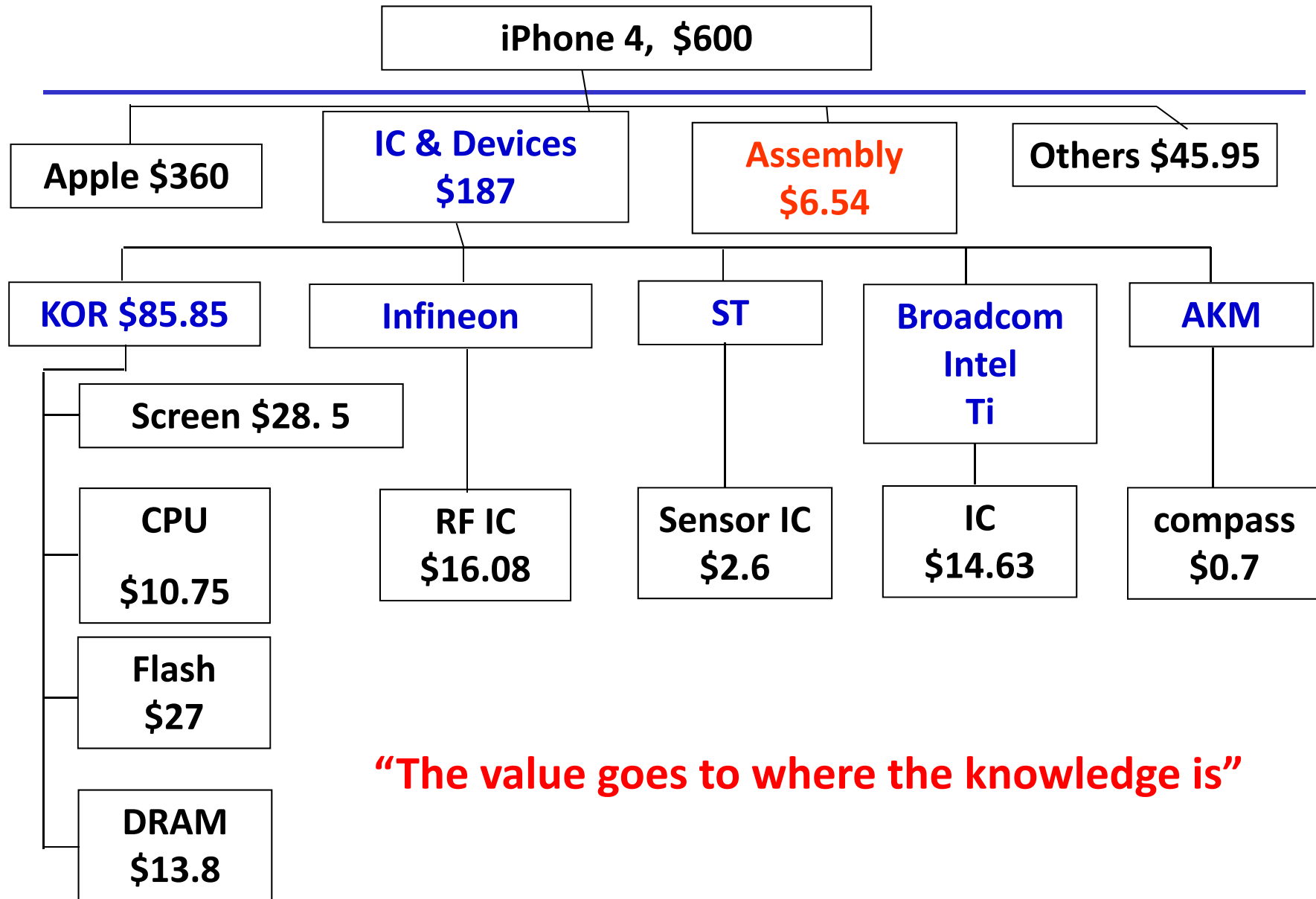


Inside iPhone



Apple 6 plus

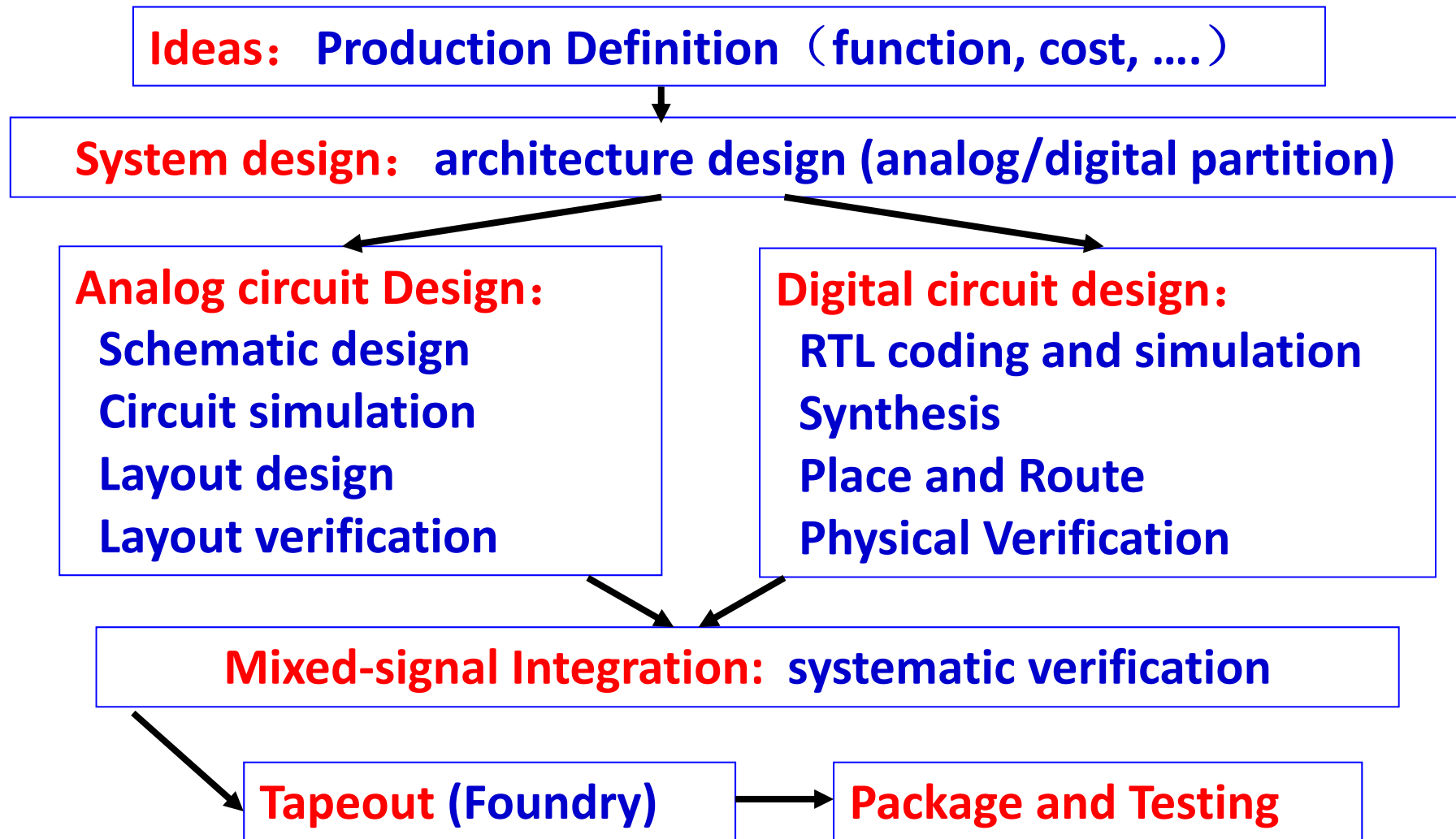




“The value goes to where the knowledge is”



IC production flow



Business Mode of IC Industry

- **Unique Business Mode:**
 - Separation of **Design** and **Fabrication**, Why?
 - Huge cost of IC production line
 - **65nm 12 inch CMOS line: >\$3 billion**
 - Very few companies construct their own line
 - **Intel, SAMSUNG, AMD, TI... we call them “IDM”.
(Integrated Device Manufacturer)**
 - Others are “Fabless”
 - Famous Foundries:
 - **TSMC/ UMC/ GlobalFoundries/SMIC/ X-FAB.....**

Top 10 Semiconductor Company, 2015

Intel	U.S.A	50.305 (Billion \$)
Samsung	South Korea	41.606
TSMC	Taiwan China	26.562 Foundry
SK Hynix	South Korea	16.917
Qualcomm	U.S.A	15.632 Fabless
Micro	U.S.A	14.816
Ti	U.S.A	12.112
Toshiba	Japan	9.734
Broadcom	U.S.A	8.421 Fabless
Avago	Singapore	6.961 Fabless

Top 20: US 8, Japan 3, EU 3, Taiwan 3, Korea 2, Singapore 1.



Top 10 Semiconductor Company, 2020

Samsung	South Korea	208.5 billion \$
Intel	U.S.A	71.9
SK Hynix	South Korea	35.27
TSMC	Taiwan China	35
Micron	U.S.A	30.9
Qualcomm	U.S.A	24.3
Broadcomm	U.S.A	20.85
Texas Instruments	U.S.A	14.38
Toshiba	U.S.A	12.3
Nvidia	U.S.A	11.72



Top 10 fabless, 2015

Qualcomm	U.S.A
Broadcom	U.S.A
Avago	Singapore
MediaTek	Taiwan China
Nvidia	U.S.A
AMD	U.S.A
HiSilicon	Mainland China
Apple	U.S.A
Marvell	U.S.A
Xilinx	U.S.A

Top 10 fabless, 2017

Qualcomm	U.S.A
Broadcom	U.S.A
Nvidia	U.S.A
MediaTek	Taiwan China
Apple	U.S.A
AMD	U.S.A
HiSilicon	Mainland China
Xilinx	U.S.A
Marvell	U.S.A
Unigroup	Mainland China

Top 10 Semiconductor Company, Q1- 2020

Intel	U.S.A
Samsung	South Korea
TSMC	Taiwan, China
SK Hynix	South Korea
Micron	U.S.A
Qualcomm	U.S.A
Broadcomm	U.S.A
Nvidia	U.S.A
Texas Instruments	U.S.A
HiSilicon	Mainland, China

More details of the famous companies in 2021

Semiconductor Companies with >\$10 Billion in Sales in 2021F

2021F Rank	2020 Rank	Company	Headquarters	2020 Total IC	2020 Total O-S-D	2020 Total Semi	2021F Total IC	2021F Total O-S-D	2021F Total Semi	2021/2020 % Change
1	2	Samsung	South Korea	58,555	3,298	61,853	78,850	4,235	83,085	34%
2	1	Intel	U.S.	76,328	0	76,328	75,550	0	75,550	-1%
3	3	TSMC (1)	China Taiwan	45,572	0	45,572	56,633	0	56,633	24%
4	4	SK Hynix	South Korea	26,094	981	27,075	35,628	1,639	37,267	38%
5	5	Micron	U.S.	22,542	0	22,542	30,087	0	30,087	33%
6	6	Qualcomm (2)	U.S.	19,357	0	19,357	29,136	0	29,136	51%
7	8	Nvidia (2)	U.S.	14,659	0	14,659	23,026	0	23,026	57%
8	7	Broadcom Inc. (2)	U.S.	15,941	1,803	17,744	18,864	2,099	20,963	18%
9	12	MediaTek (2)	Taiwan	10,985	0	10,985	17,551	0	17,551	60%
10	9	TI	U.S.	12,731	843	13,574	15,889	1,015	16,904	25%
11	15	AMD (2)	U.S.	9,763	0	9,763	16,108	0	16,108	65%
12	11	Infineon	Europe	7,542	3,683	11,225	9,113	4,503	13,616	21%
13	10	Apple* (2)	U.S.	11,440	0	11,440	13,430	0	13,430	17%
14	14	ST	Europe	6,804	3,374	10,178	8,400	4,174	12,574	24%
15	13	Kioxia	Japan	10,553	0	10,553	12,132	0	12,132	15%
16	17	NXP	Europe	7,582	809	8,391	9,711	1,004	10,715	28%
17	19	Analog Devices (3)	U.S.	7,722	405	8,127	9,575	504	10,079	24%
—	—	Top-25 Total		364,170	15,196	379,366	459,683	19,173	478,856	26%

(1) Foundry

(2) Fabless

(3) Includes acquired company's sales in 2020 and 2021 results.

Source: Company reports, IC Insights

*Custom devices for internal use.

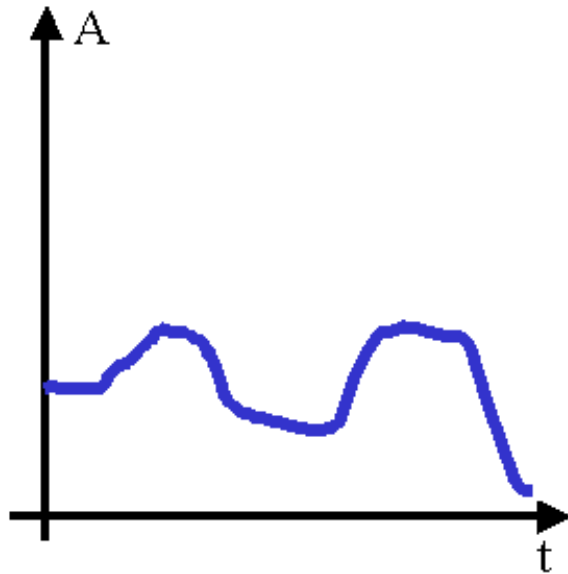
What you can do in IC industry

- System engineer
 - audio and video signal processing, communication...
- IC design
 - Analog IC design : ADC/DAC/PLL/Power/....
 - Digital IC design:
 - Front end: RTL coding
 - Backend: Synthesis, place and route, timing analysis
- Foundry
 - Device modeling, processing development
- EDA Tool: Cadence/Synopsys/Mentor
- IC Package and IC testing

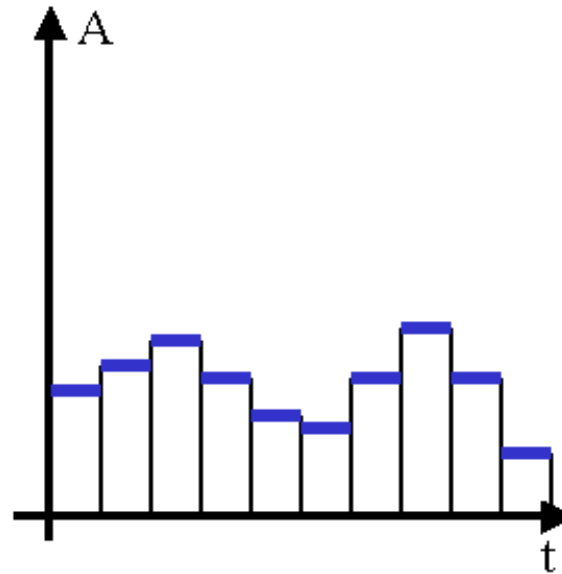
Differences between Analog and Digital (1)

- Signal representation
 - Digital
 - Information is recorded both in discrete time and discrete amplitude
 - Only have two status: supply voltage (“1”) or ground (“0”).
 - Often use multi-bit bus to express a signal
 - Analog
 - Information is recorded both in continuous time and continuous amplitude
 - The value at any time is important
 - Single-ended or differential

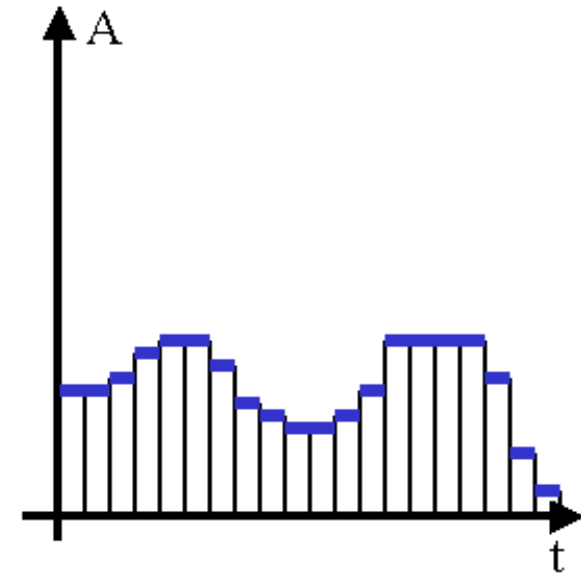
Signal representation: A vs D



Analog signal –
continuously varying



Digital signal – large
time divisions



Digital signal – small
time divisions

Differences between A/D : Design methodology

– Digital

- Circuit is described with hardware description language (HDL)
- Basic building block is standard cells (INV, NOR..)
- Layout is obtained by software (automated)
- Tools play very important role in the design flow

– Analog

- Circuit is described with schematic or netlist
- Basic devices are transistors, capacitors and resistors, Higher-level extraction is amplifier, filter, comparator...
- Layout is designed manually. “Art of layout design”
- Wisdom and experience of human predominate the design

A full_adder example

```
module add (co, s, a, b, c)
```

```
input a, b, c ;
```

```
output co, s ;
```

```
    xor (n1, a, b) ;  
    xor (s, n1, c) ;  
    nand (n2, a, b) ;  
    nand (n3, n1, c) ;  
    nand (co, n3, n2) ;
```

```
endmodule
```

=

Always

```
@(a or b or c) begin
```

```
    s = ( a^b^c);
```

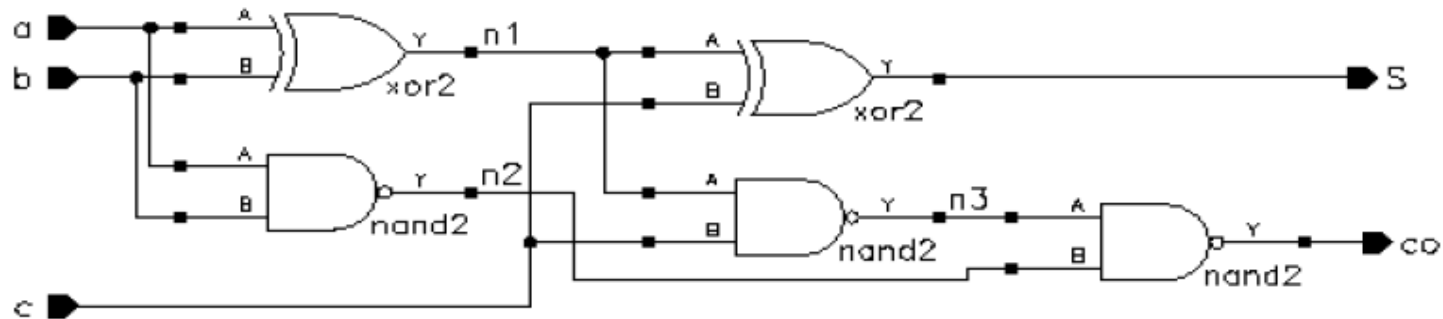
```
    T1=a&b;
```

```
    T2=a&c;
```

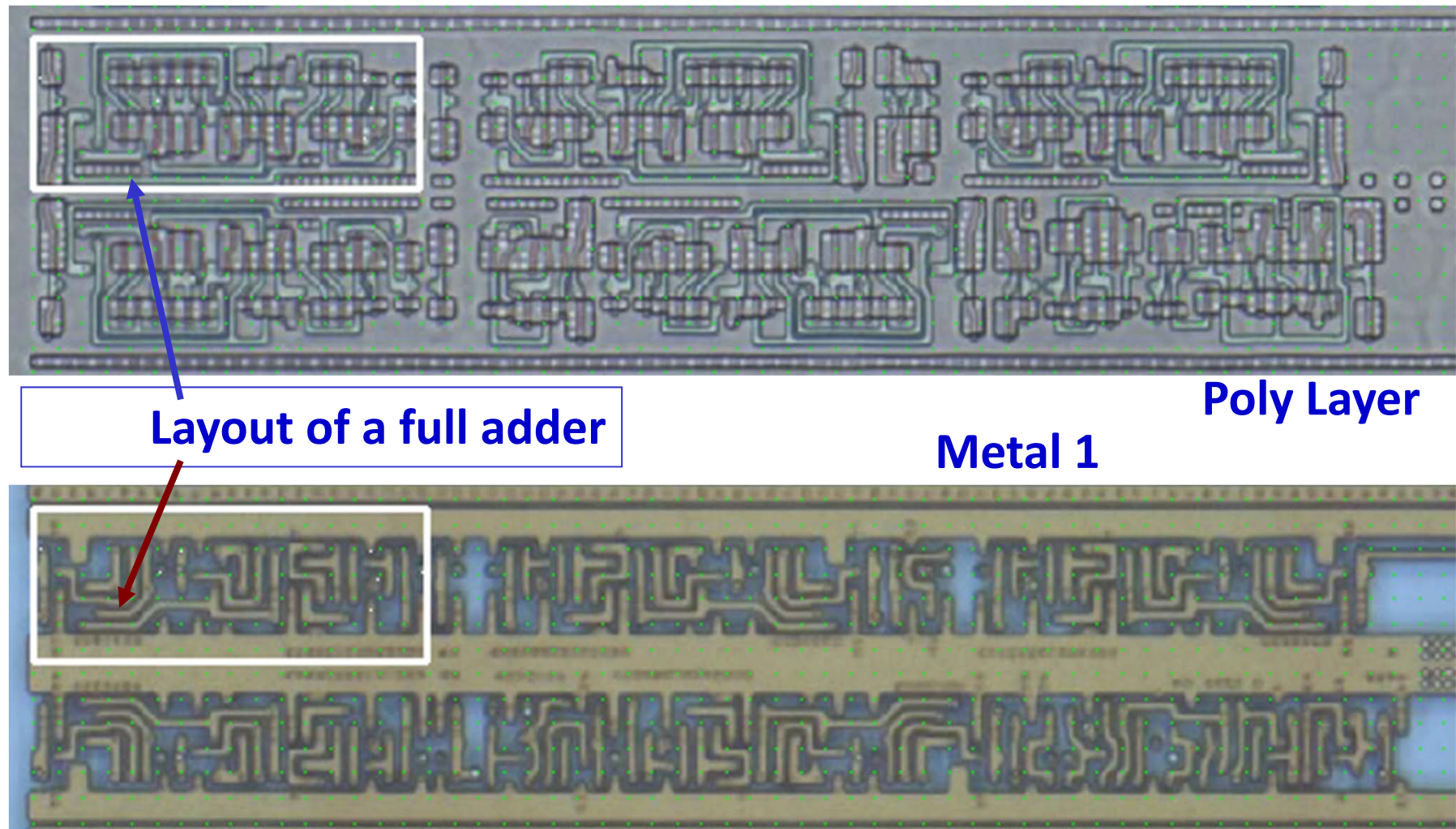
```
    T3=b&c;
```

```
    co= (T1 | T2) | T3;
```

```
end
```

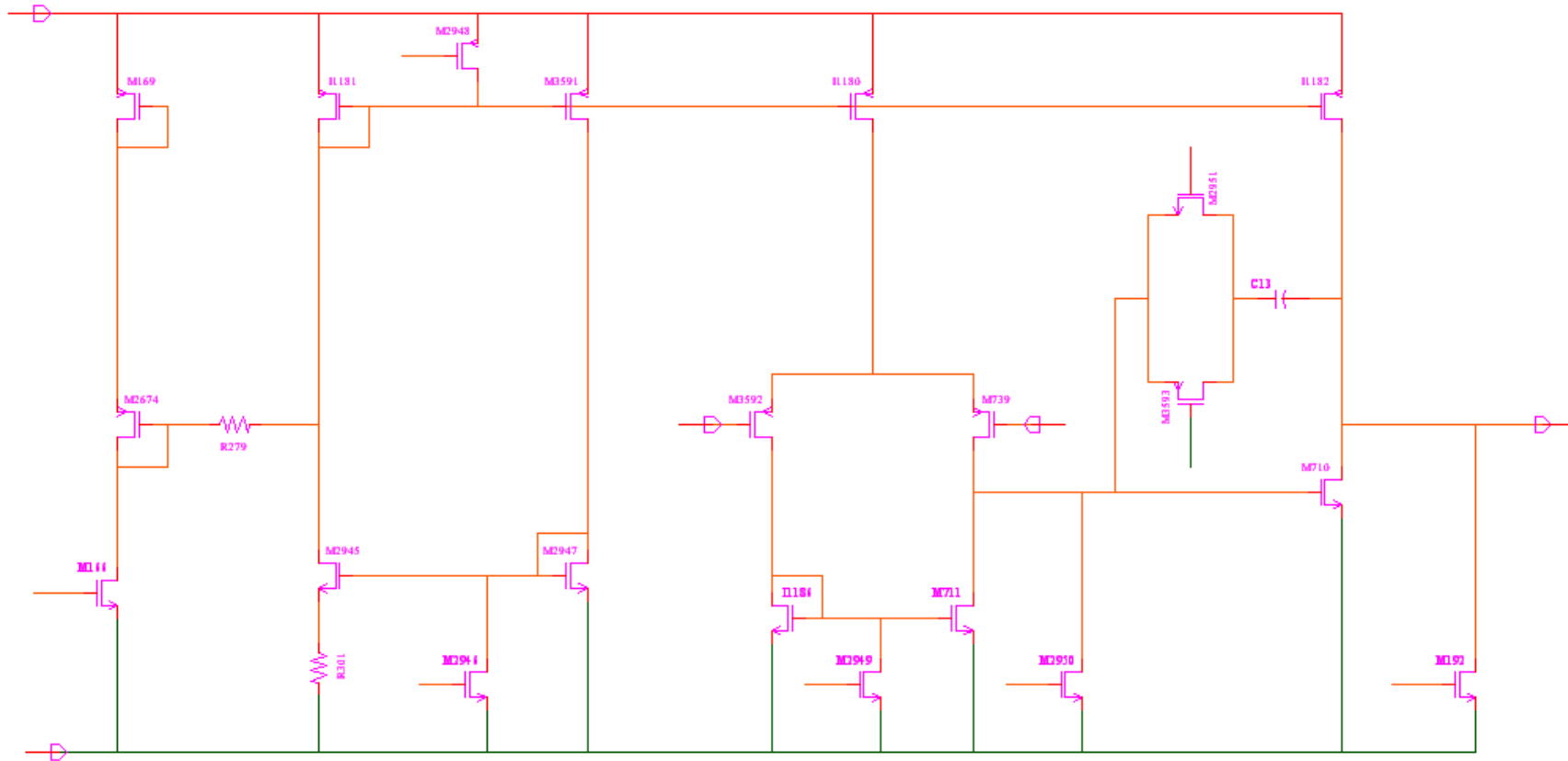


Full adder used in a digital circuit



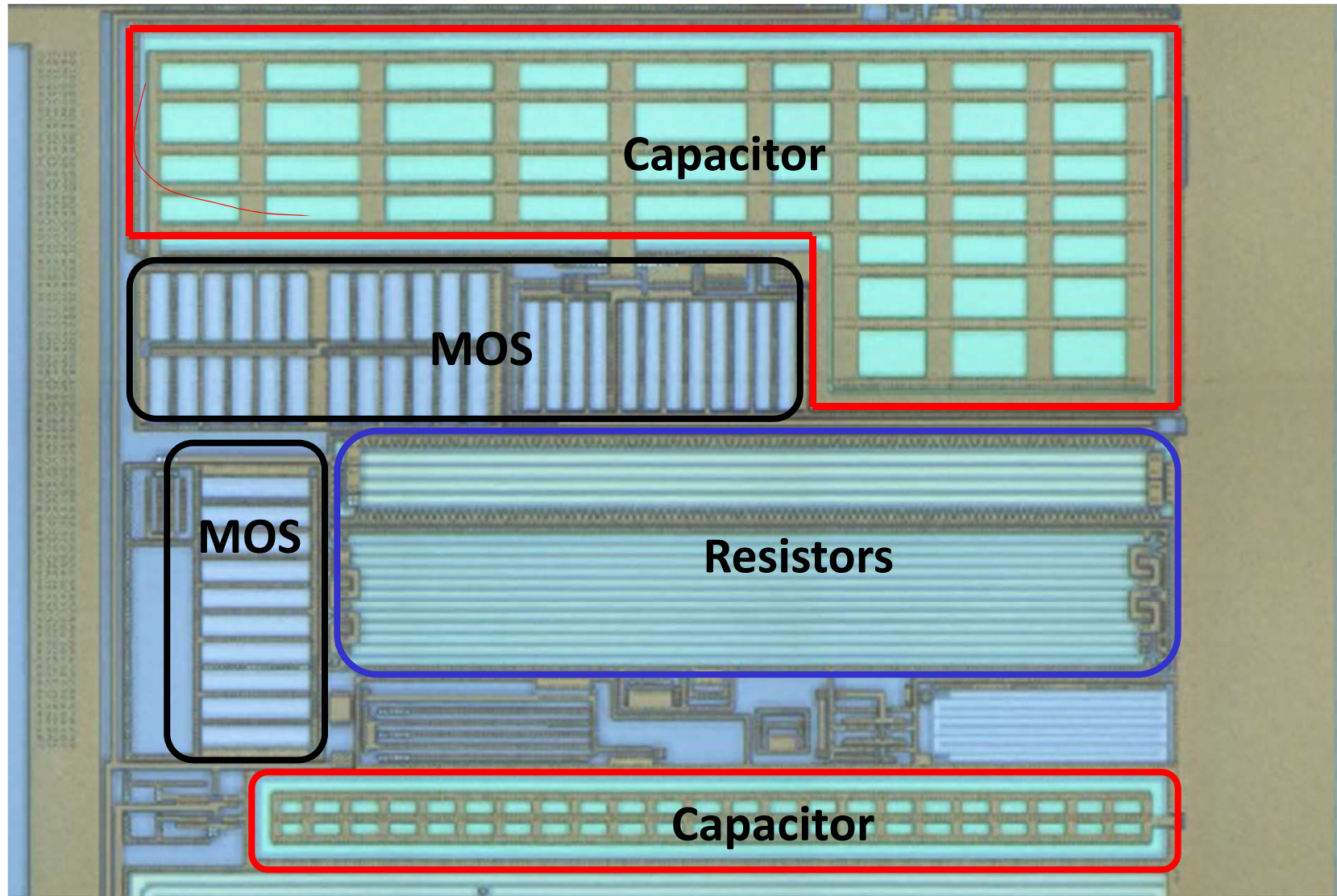
- The layout is obtained by a auto-place and route software

An Operational Amplifier Example



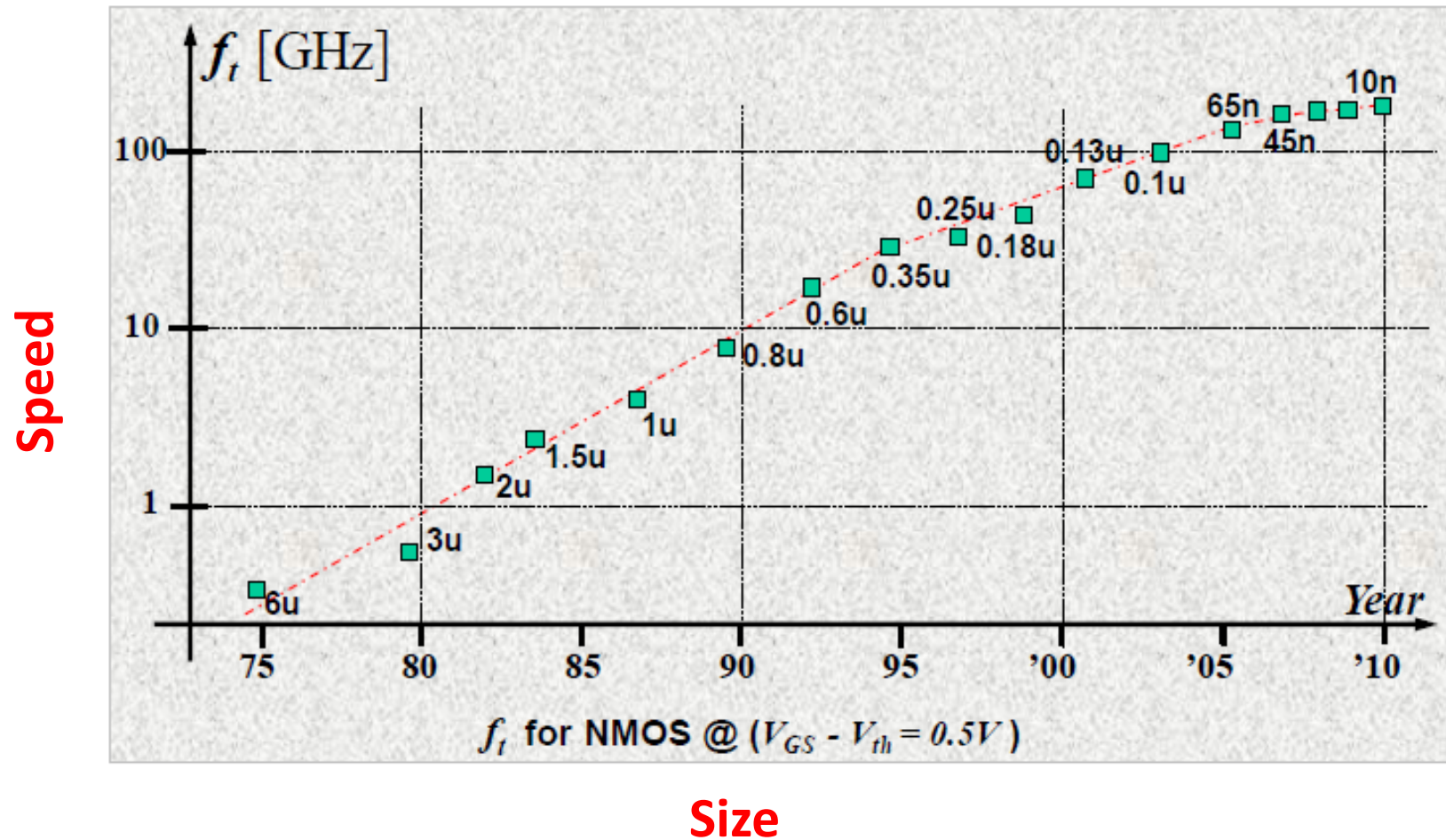
- The size of each transistor is designed elaborately to meet the performance requirement

Die photo of the OP



Fabrication capability roadmap

CMOS Technology Evolution versus Time

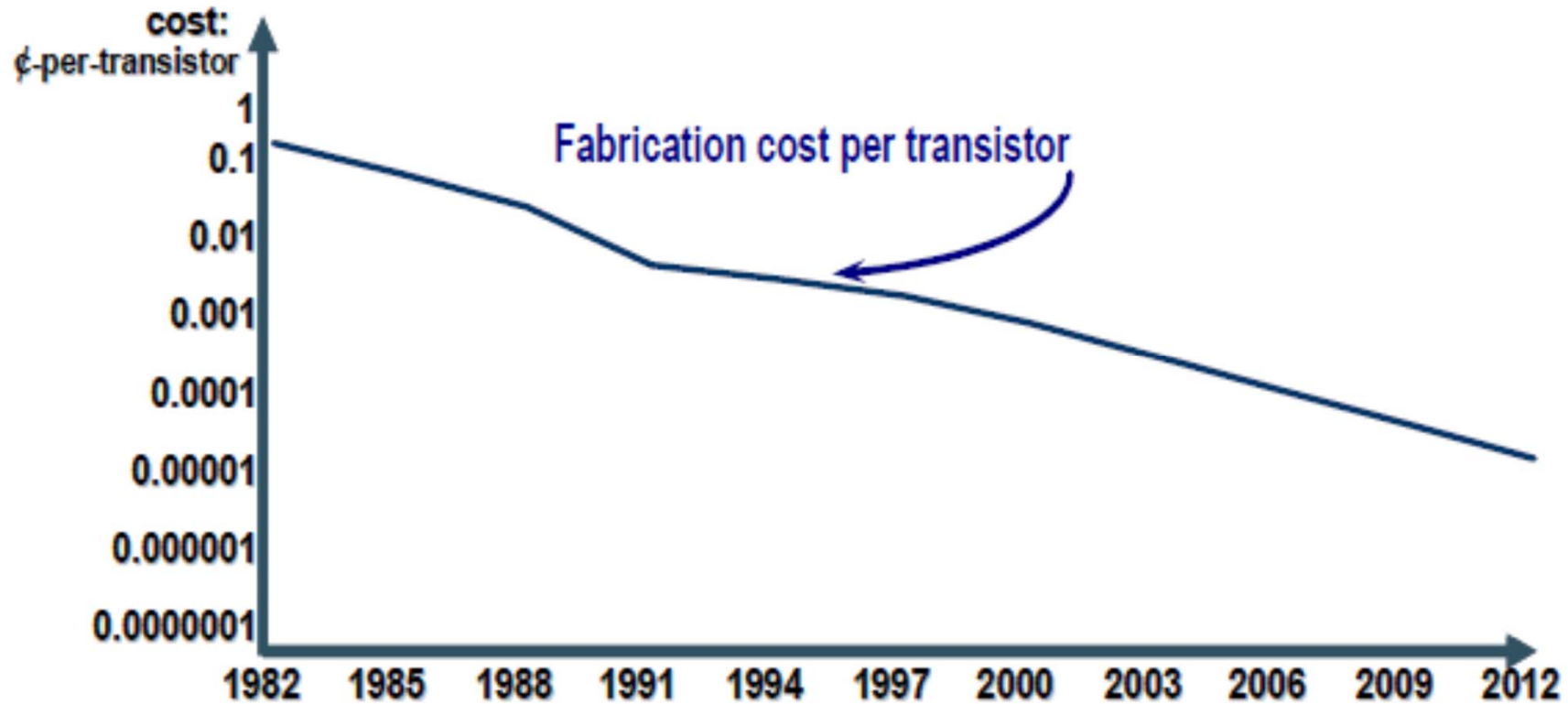


Intel's technology

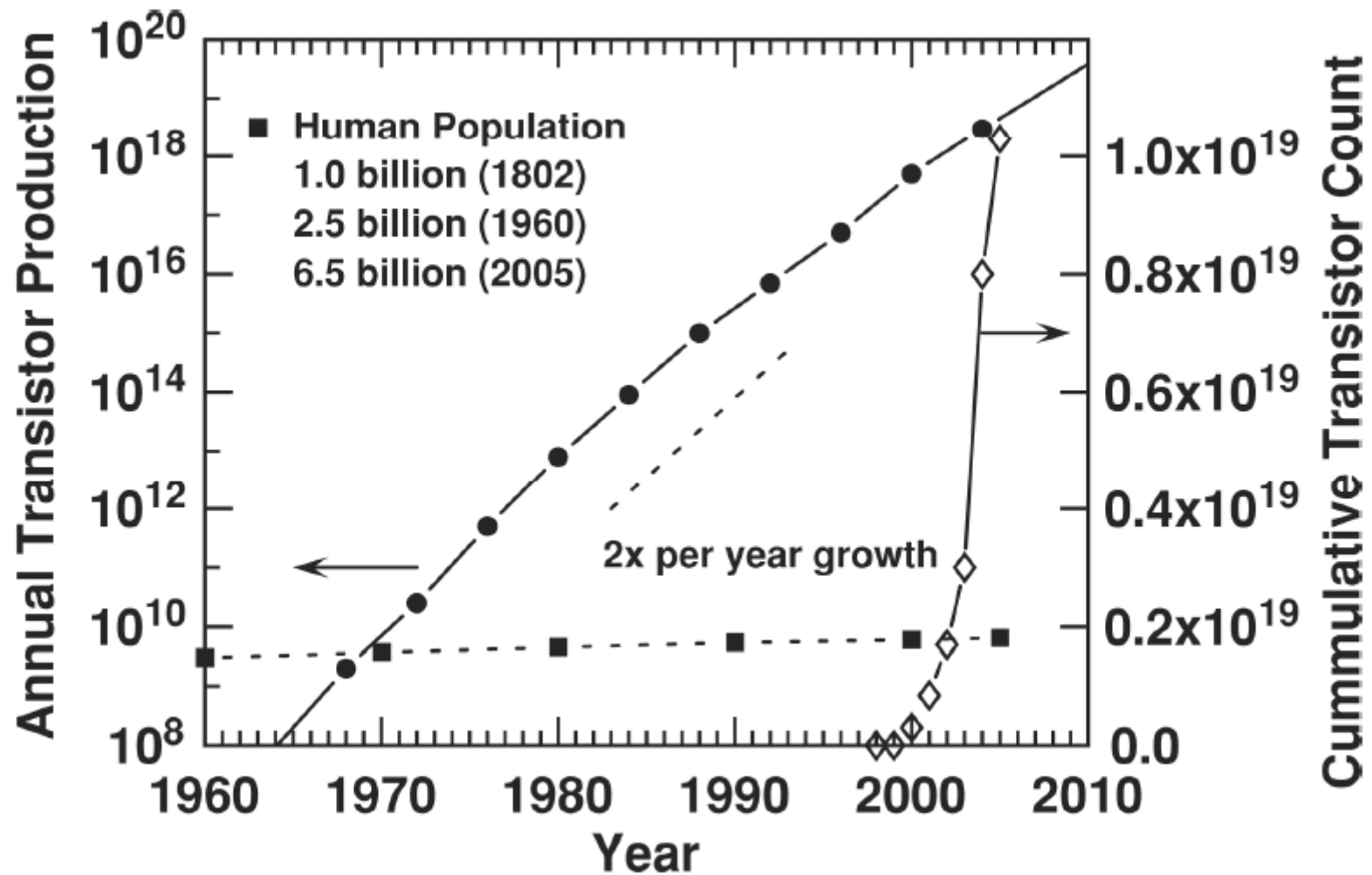


Benefits of process shrinking

- Cost/function decreasing



1.5 billion transistors / person



IC industry is driven mainly by **digital** CMOS technology

- Digital circuit: Fully benefited From IC device scaling
 - Cost/function decreased by 29% each year
 - CPU, memory, DSP
- Analog circuit: not fully benefited
 - Device scaling mandates drop in supply voltage
 - Device matching difficulties
 - **Cost/function almost constant of increase**
- More and more signal processing functions are shifted from analog domain to digital domain
 - **This is a digital world!**

Comparisons of analog/digital products

- 1990's, the 1st- generation pure analog mobile phone
- Before 2007, 2.5G digital phone
- The past years: Smart phone



Will analog circuit disappear in the future?

- **Definitely No!** Why?
- Currently, ASIC flow can only synthesize circuits with “medium frequency” (below 500MHz)
 - Standard cells and I/Os are designed in transistor level
 - Memory cell designed and simulated by “analog” method
- To improve the performances of the current digital circuit, many “analog” problems need to be solved (increase speed, reduce power dissipation)
 - Charge sharing
 - Interconnect parasitic
 -

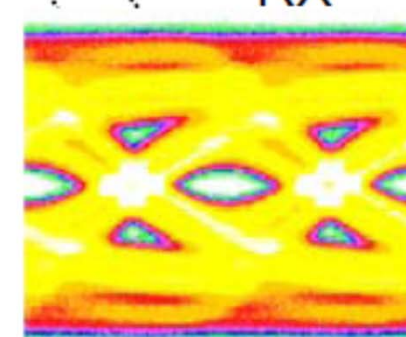
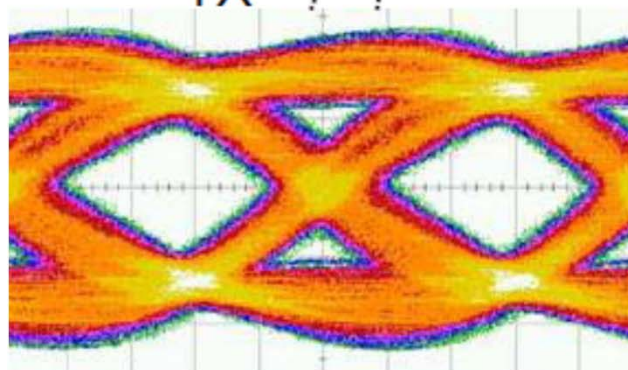
Example: digital signals on PCB

- High speed digital signals are distorted by parasitic RC
- Received signal is “0” or “1”?
- Analog circuits are need to assist signal receiving
 - LVDS
 - CML
 - SerDes



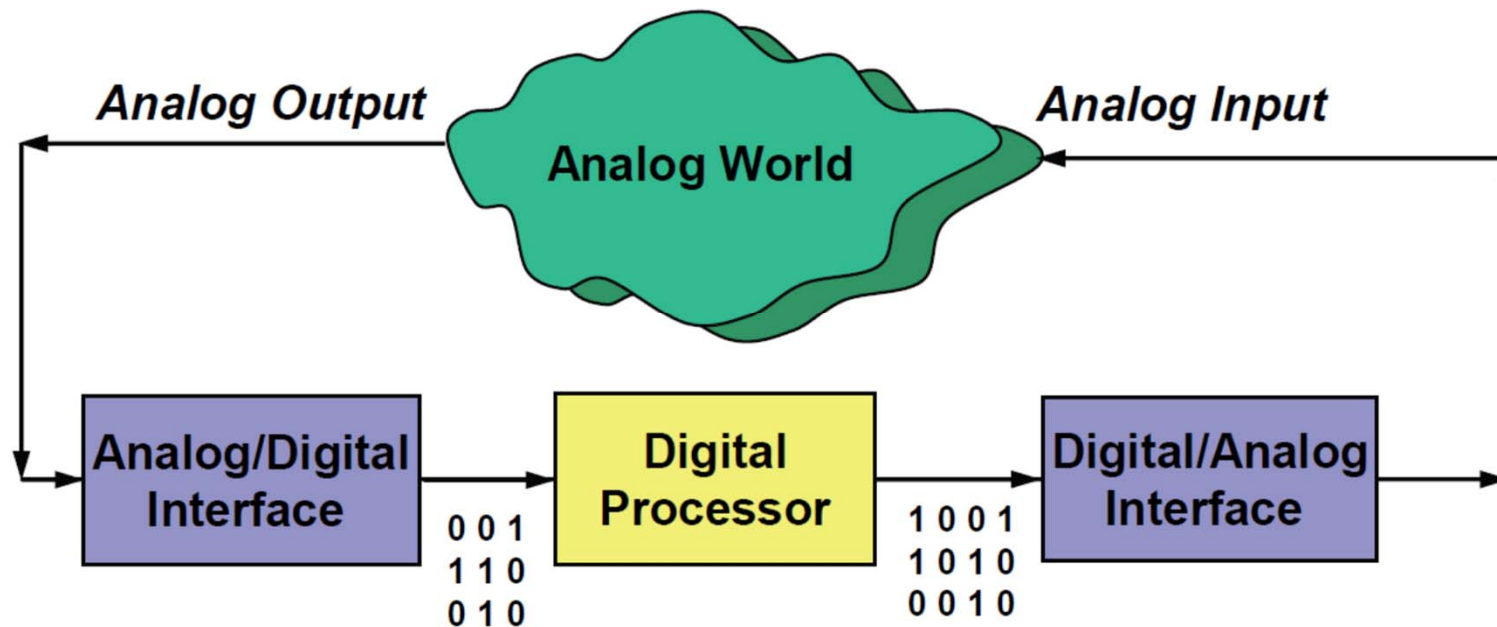
TX

RX

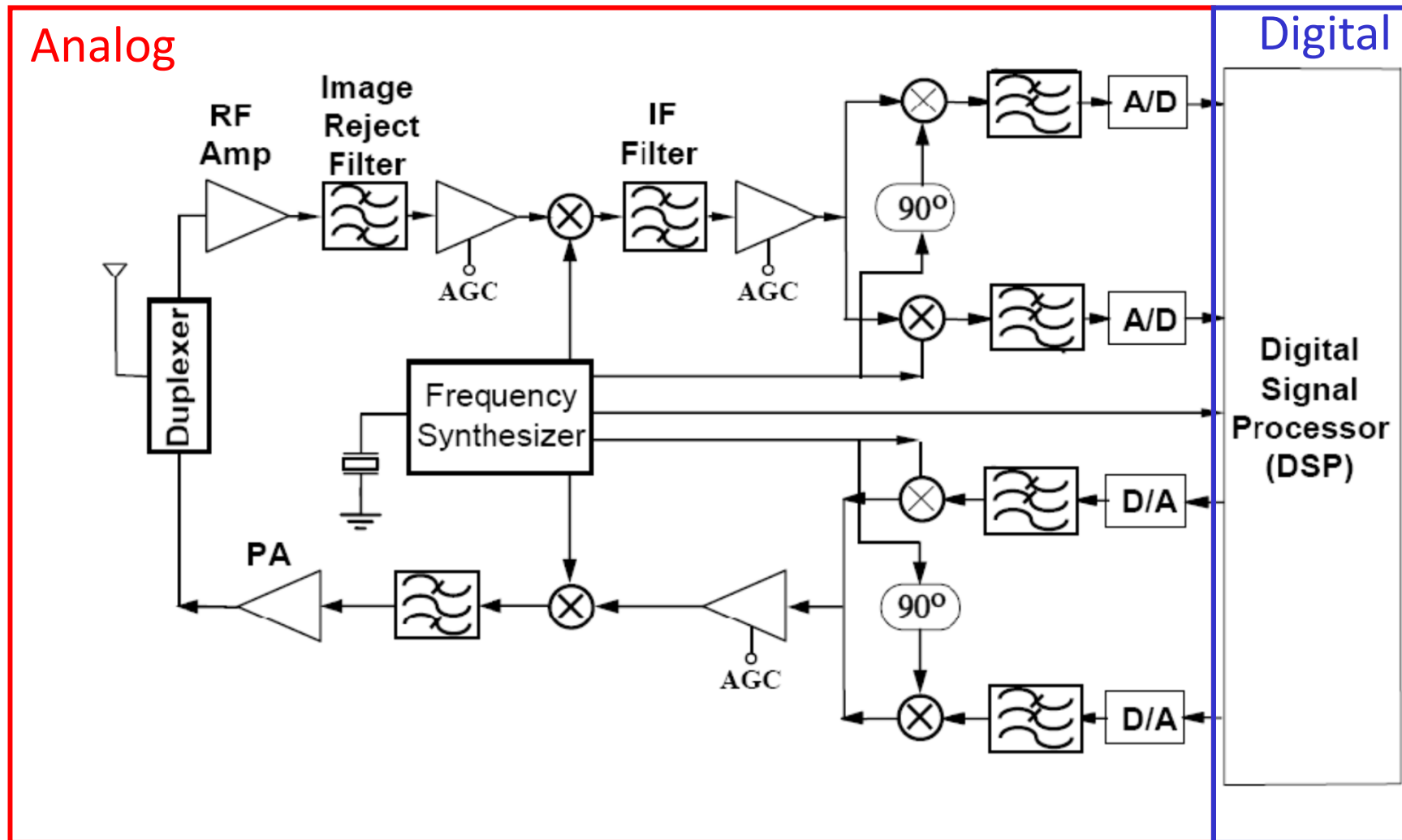


The truth behind why analog will always exist

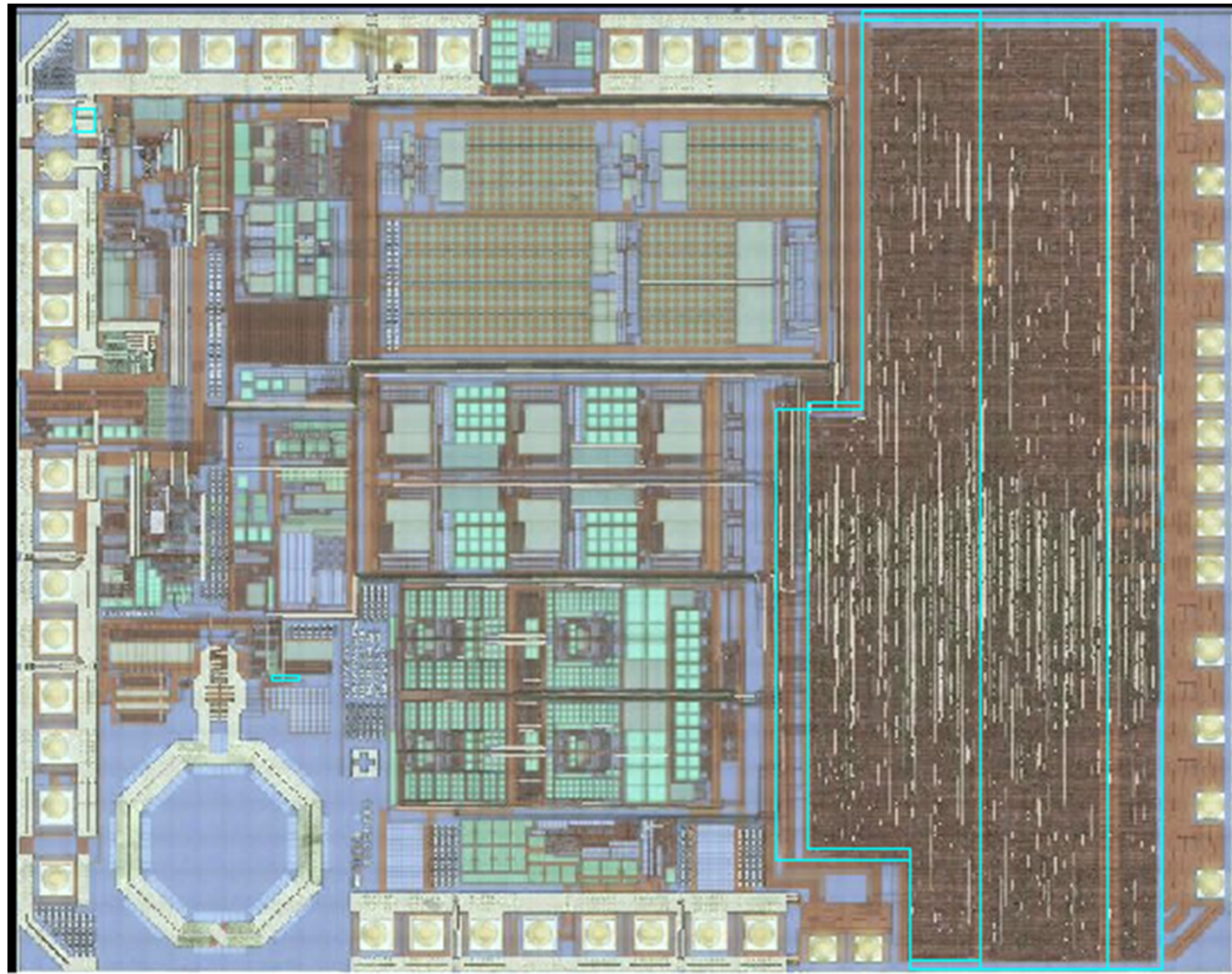
- Every thing in IC is **analog!**
- Signals in real world are analog. Interface circuits must be used to convert them into digital
 - For most cases, analog paths are critical paths



Example (1): An RF transceiver for mobile phones



Chip photo



Analog

digital



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XI'AN JIAOTONG UNIVERSITY

Analog modules in a mobile phone

- Filters and ADC/DAC



Contains in integrated form:

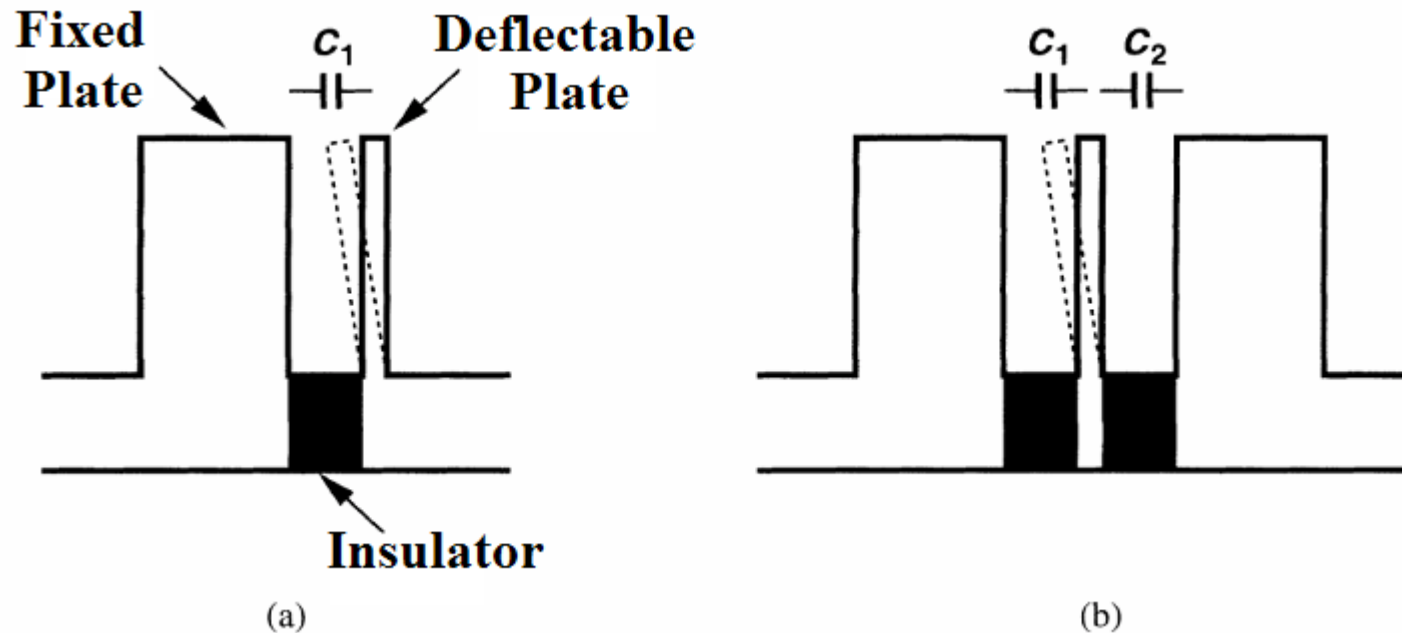
- 4 Rx filters
 - 4 Tx filters
 - 4 Rx ADCs
 - 4 Tx DACs
 - 3 Auxiliary ADCs
 - 8 Auxiliary DACs
- Dual Standard, I/Q
- Audio, Tx/Rx power control, Battery charge control, display, ...

Total: Filters → 8

ADCs → 7

DACs → 12

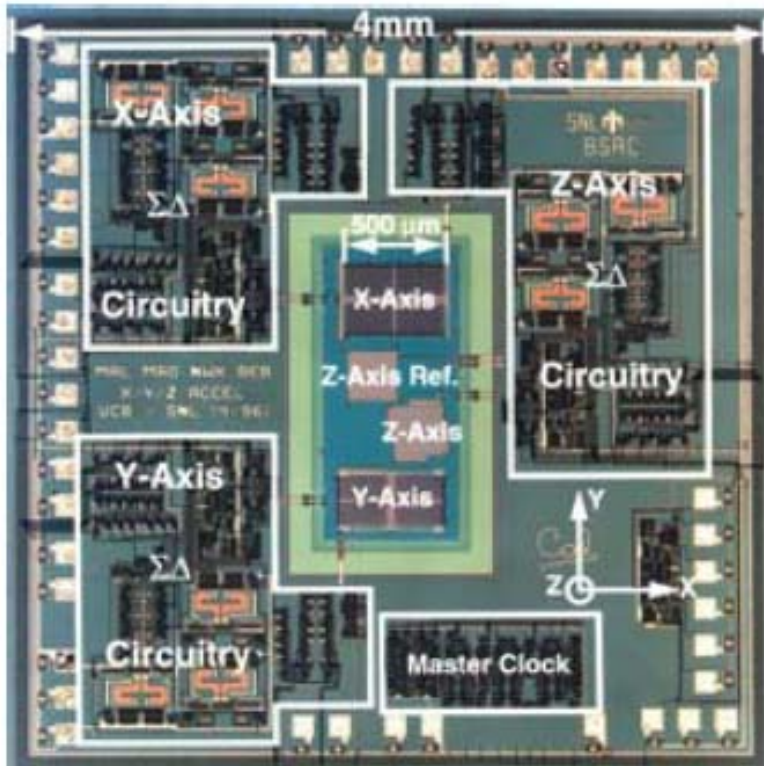
Example (2): Sensors (accelerometer)



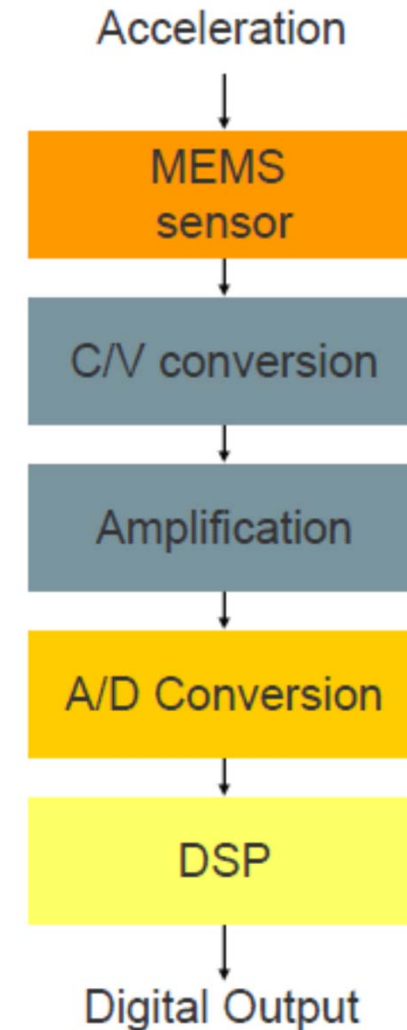
(a) simple accelerometer

(b) differential accelerometer

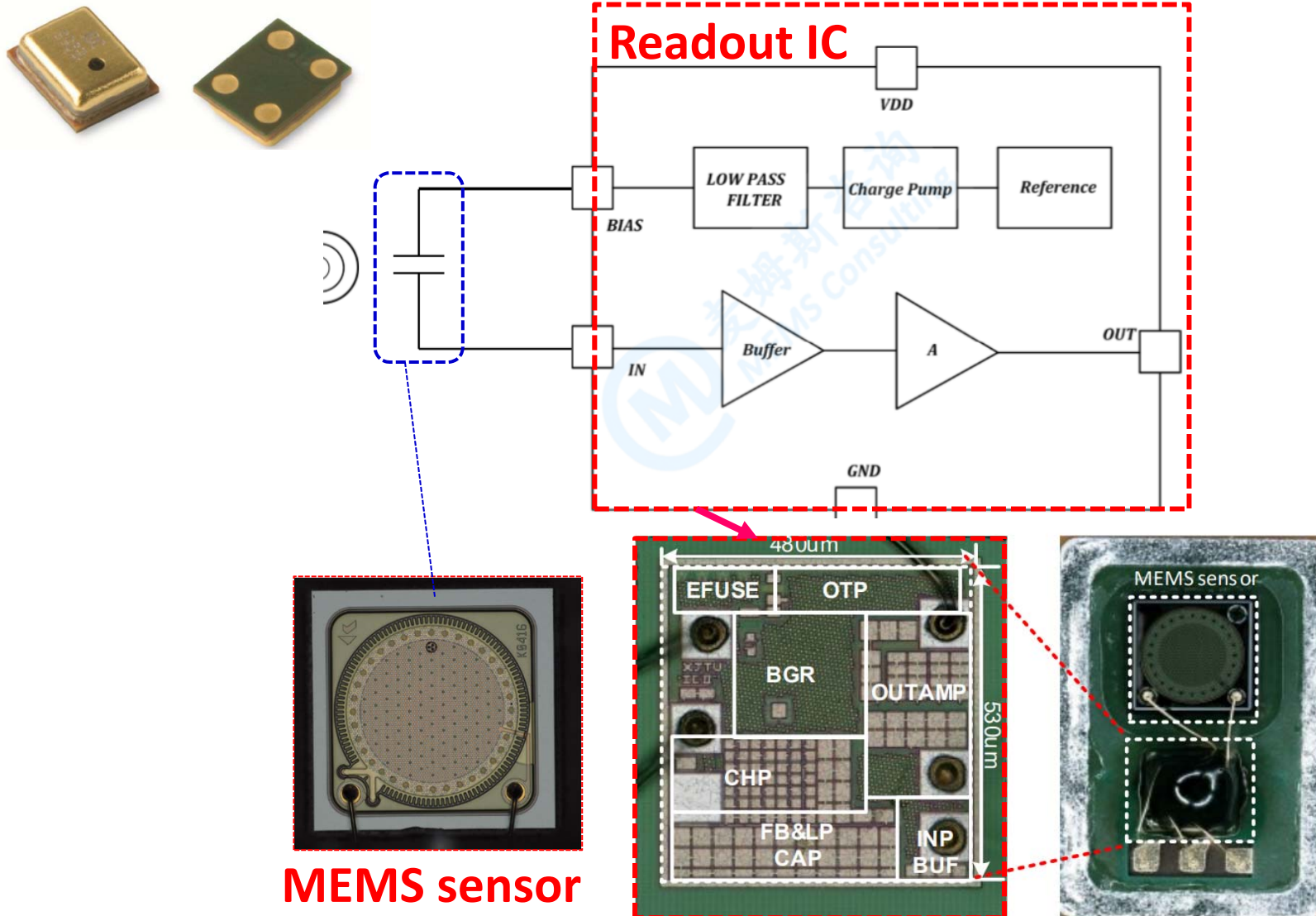
Example (2): MEMS Accelerometer



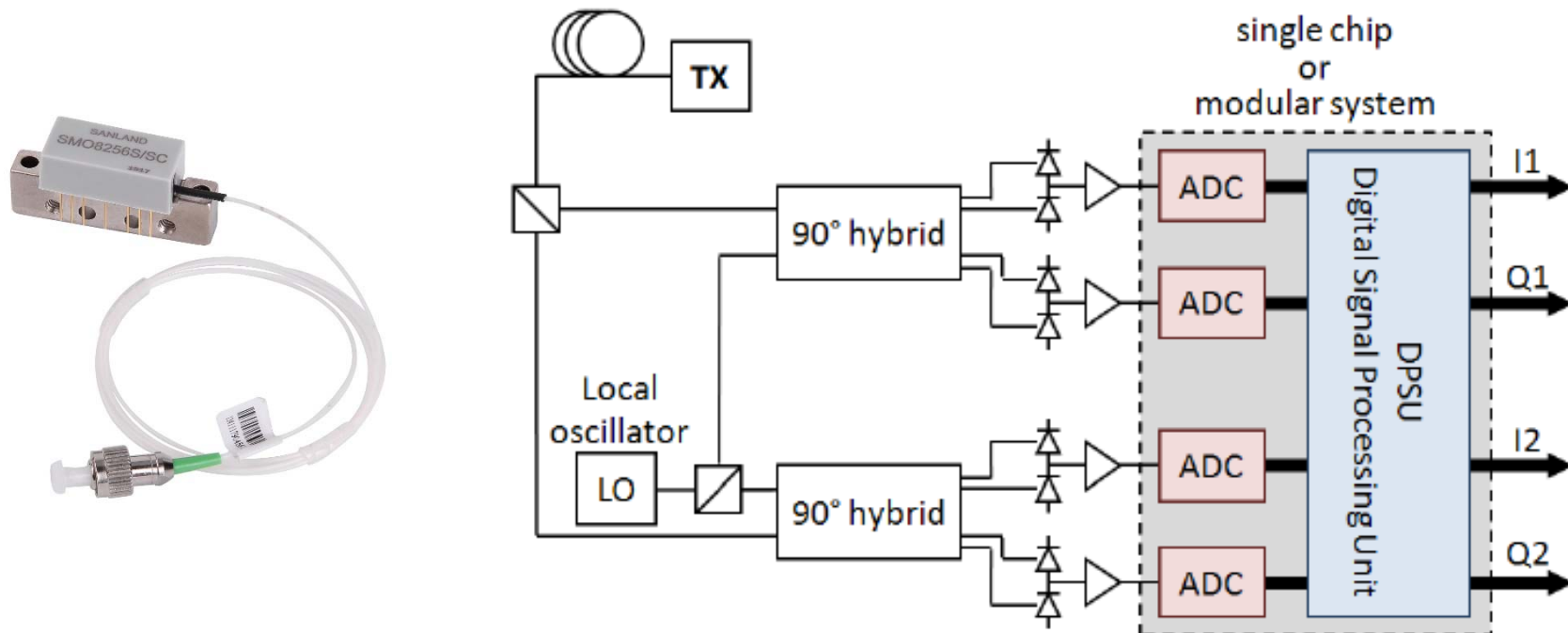
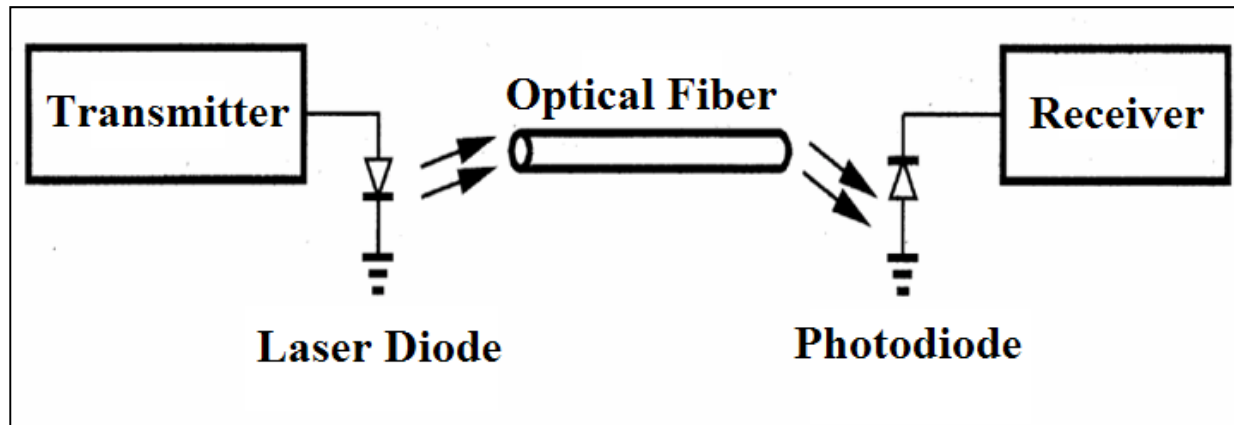
M. Lemkin and B. E. Boser, "A Three-Axis Micromachined Accelerometer with a CMOS Position-Sense Interface and Digital Offset-Trim Electronics," *IEEE J. Solid-State Circuits*, vol. SC-34, pp. 456-468, April 1999



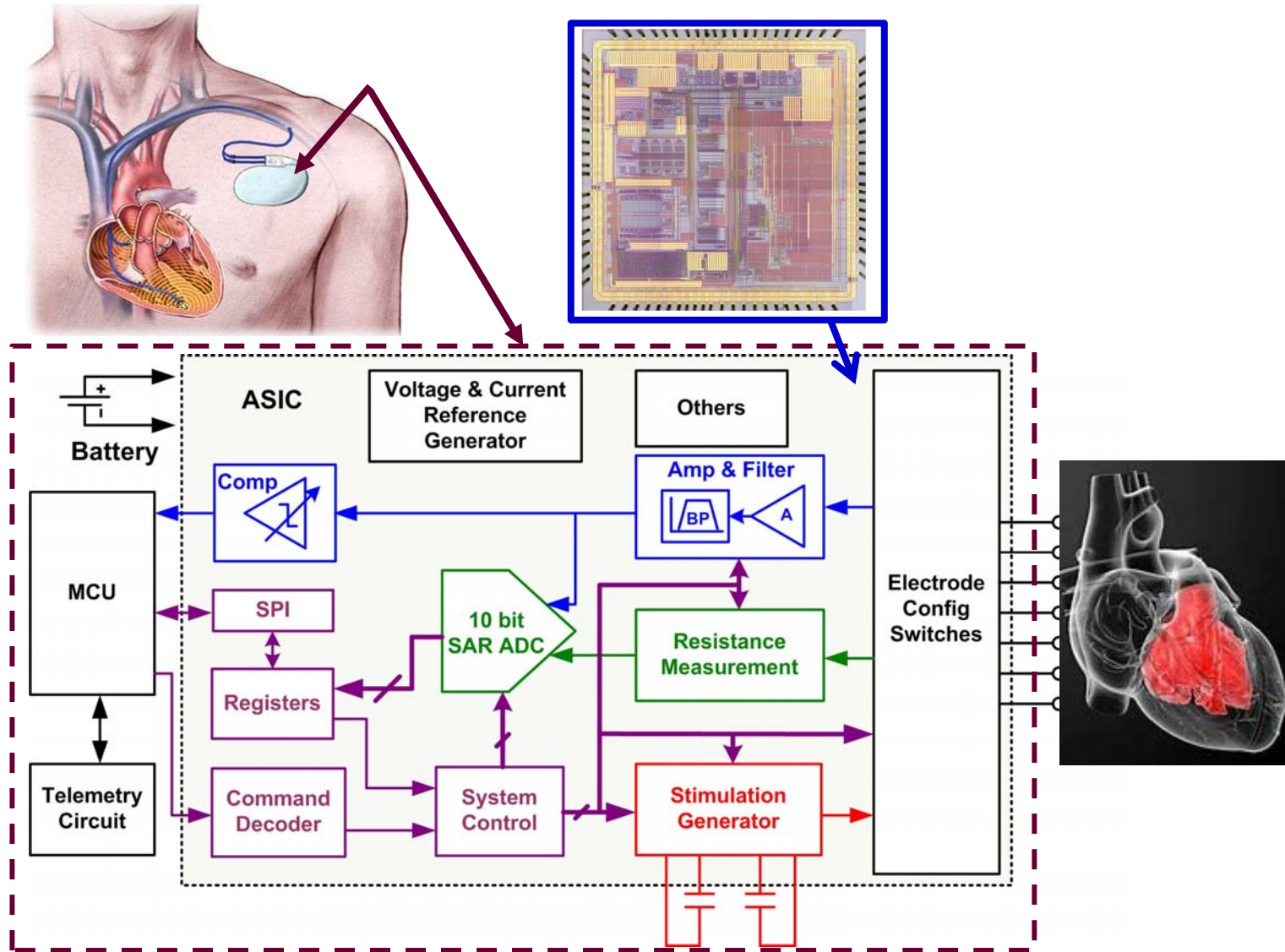
Example (3): MEMS Microphone



Example (4): Optical Receivers



Example (5): Biomedical (cardiac pacemaker)

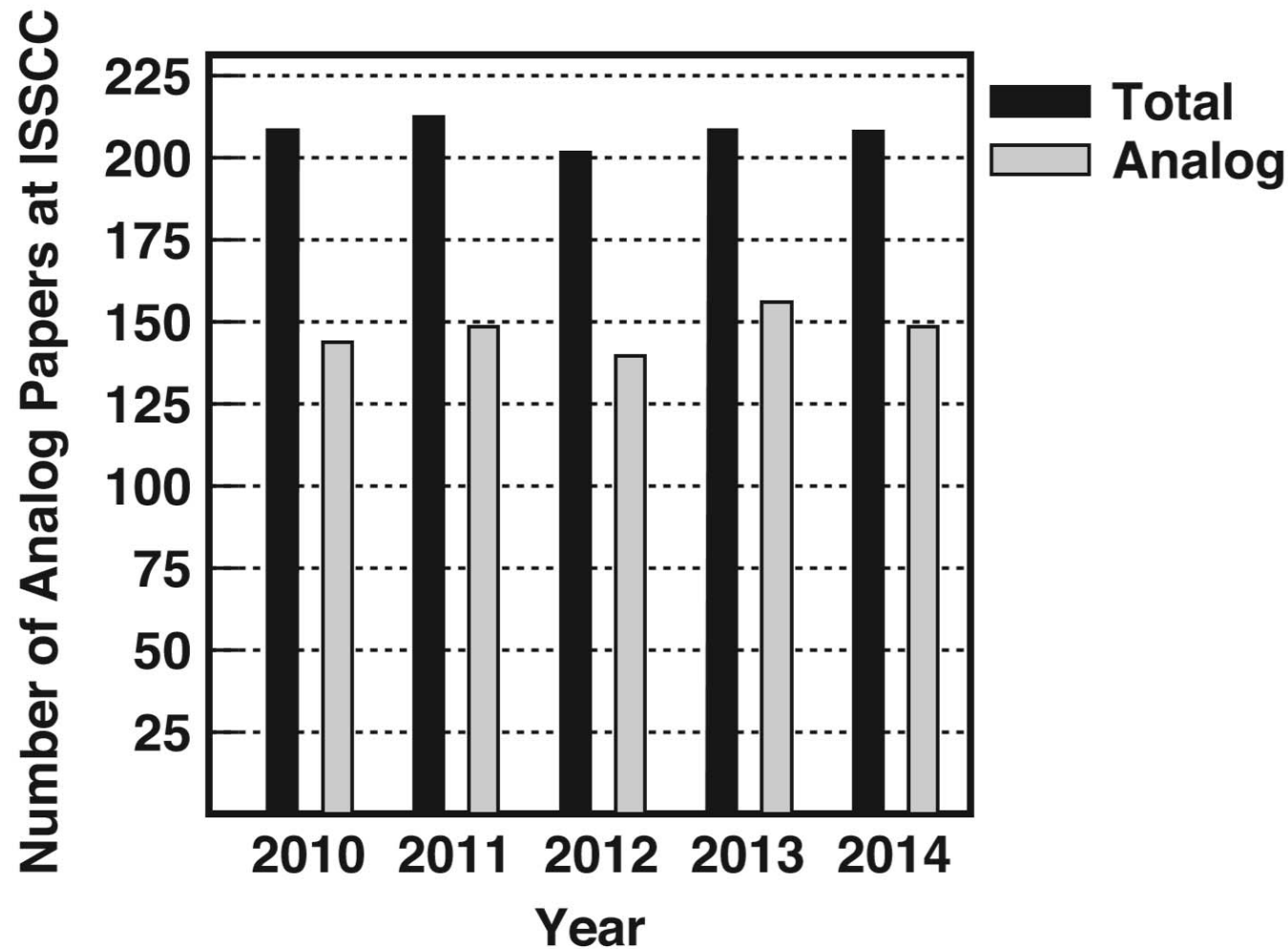


Why analog design? (summary)

- Interface between analog signal and digital signal
(ADC)
- Amplifying and processing for natural signals
(Amplifier & Filter)
- Design in higher speed digital circuits.
(High-speed digital design is in fact analog design.)
- Mixed-signal circuits and SOC
- More important:
 - Salary of a analog engineer is often 1.5 ~ twice of a digital engineer

Analog IC Design in Research Field

- ISSCC is dominated by analog papers



Important analog modules

- What can we design :
 - Power Management: AC-DC, DC-DC, LDO
 - RF modules: LNA, Mixer, PA, Synthesizer
 - IF modules: filter, VGA
 - Interface modules: ADC, DAC
 - Clock signal generation: PLL

Why is analog design difficult?

1. Analog circuits deal with **multi-dimensional tradeoff** of speed, power, gain, precision, supply, ...
(Digital circuits deal primarily with **speed power tradeoff**.)
2. Due to speed and precision requirements, analog circuits are much more **sensitive** to noise, crosstalk, and other interferers.

Why is analog design difficult? (cont.)

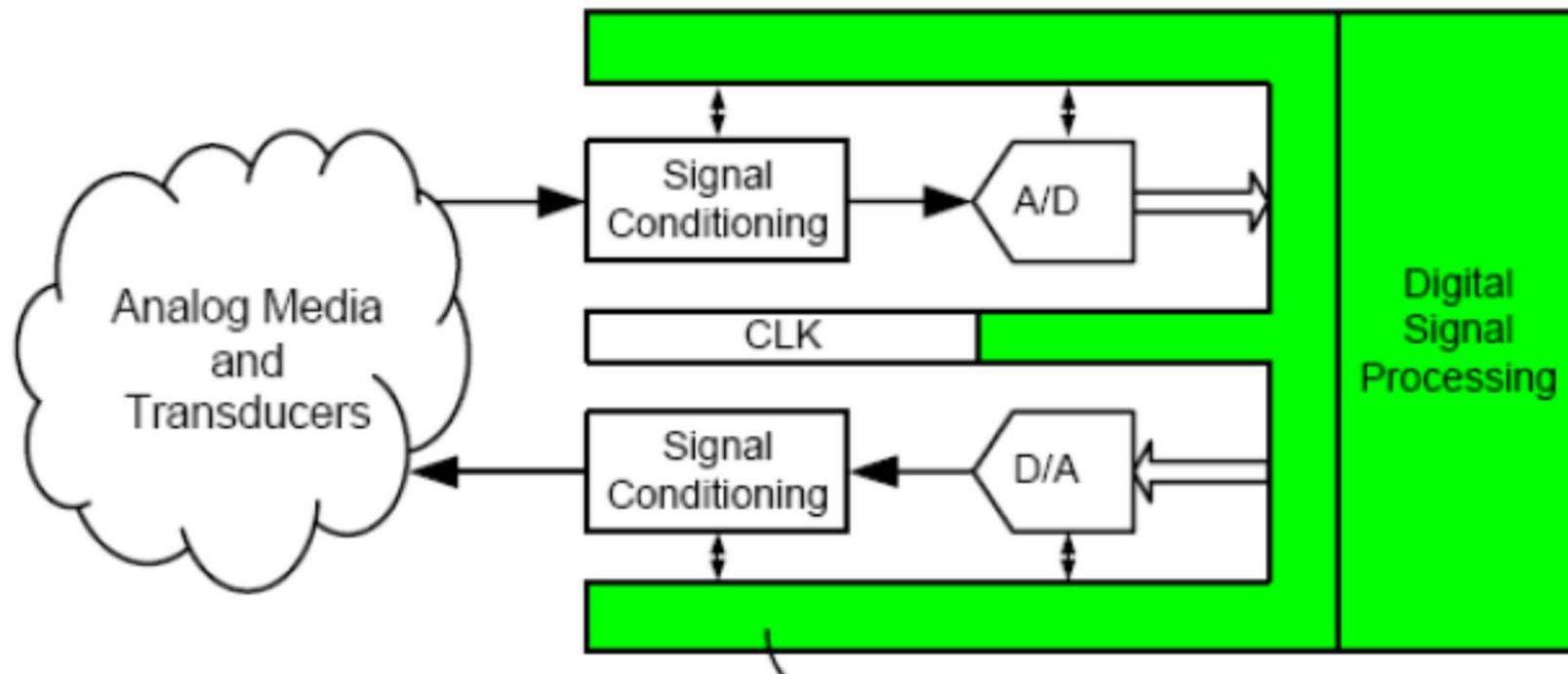
3. Analog circuits are much more sensitive to **second-order device effects**.
4. High performance analog circuit design can rarely be automated - typically require **hand-crafted design and layout**.
5. Modeling and simulation of analog circuits is still problematic, requiring **experience and intuition**.

Why is analog design difficult? (cont.)

6. Economic forces require the development of analog circuits in **mainstream digital processes**.
7. Economic forces pushing the integration of analog and digital functions onto a **single substrate (SOC)**.
8. Continuous reduction of feature sizes, the power dissipation and supply voltages (**technology scaling**) further exacerbates analog design challenges.

Where to go?

- Digitally assisted analog design
 - Mixed-signal IC design



The future of IC industry

- More Moore?
 - Develop transistors with length less than 5 nm.
 - CPUs, memories, and DSP are beneficial
 - Small companies and universities are out
 - Only a few companies can afford
- More than Moore ?
 - Continue to use IC processes with large feature size (0.18 μ m, 0.13 μ m), but integrate more functions on a chip (*RF, MEMS, bio, optical...*)
 - *Good for analog IC design*
 - Small companies and universities prefer this

What will you learn in this course

- Learn basic knowledge for CMOS analog IC design, including but not limited to:
 - Chapter 2: Basic MOS Device Physics
 - Chapter 3: Single-Stage **Amplifier**
 - Chapter 4: Differential **Amplifier**
 - Chapter 5: Passive and Active Current Mirrors (**for amplifiers**)
 - Chapter 6: Frequency Response of **Amplifier**
 - Chapter 8: Feedback (**based on amplifiers**)
 - Chapter 9: Operational **Amplifier**
 - Chapter 10: Stability and Frequency Compensation (**for amplifiers**)

Journals and Conferences for analog researcher

- Journals:

- (JSSC) IEEE Journal of Solid-State Circuits
- (TCAS) IEEE Transaction on Circuit and System I ,II
- IET Electronics Letters、IEICE Trans. Electronics
- Analog Integrated Circuits and Signal Processing
- 半导体学报(Journal of Semiconductors)

- Conferences:

- ISSCC, CICC, ESSCIRC, ASSCC, VLSI
- International Symposium on Circuits and Systems (ISCAS)

Assignments- #1

- 请在互联网上搜索资料，并回答2021年中国大陆集成电路产品进口总额是多少？可能的话，列举这些进口产品的细分领域以及它们各自的进口额。
- 请借助互联网，解释什么是“瓦森纳”协定；按照自己的理解简要解释该协定对我国科技发展的影响。
- 请查阅美国ADI公司(“Analog Devices”)网站给出的产品分类，并列出模拟集成电路的产品大类的中英文对照。
- 请从附件ISSCC-2022的会议日程表中，找出所有个技术 session 的名称，列在一个表中，给出中英文对照，并标注出与模拟IC有关的session